Higher-Dimensional Consistency of Geometry and Modular Flow: Physical Source Identification and a Second-Order Extension

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October 2026

Abstract

An observer reads ball algebras at different positions, sizes and rest frames. We ask when those readings agree with one geometry. In a conformal theory on Minkowski space, the first variation of the geodesic ball's boundary area minus $(d-1)/R$ times its volume equals minus the CHM-weighted linearized Einstein tensor at every radius. The conditional law $\eta\,\delta F_B + \delta\langle K_B\rangle = 0$ is therefore equivalent to $G^{\rm lin} = (2\pi/\eta)\delta\langle T\rangle$.

A nonzero-frequency source identifies this intrinsic linear response up to normalization even when internal length scales are allowed. Local coherent and neutral scalar preparations provide smooth finite-energy Radon witnesses; a bounded pulse inside a permitted shared hull supplies a more general positive-energy test, with local source regularity still required. A scalar's physical modular energy includes a boundary charge. Its subtraction, or the matched Wald density, is essential: raw energy cannot satisfy the pure-area bulk law on all small balls.

One frame fixes only the normal constraint. Boost comparisons determine the sourced tensor, with small-rapidity stress noise amplified as β^{-2}. Free Weyl curvature is invisible at linear order. Stress-containing selected algebras give the same sourced tensor but different relative entropies. Under the supplied radial readout model, conditional multipole information falls as $e^{-4k\log k}$.

We extend the local conditional law to second order. A quadratic extrinsic-curvature correction gives the full second-order Einstein residual from all small boosted balls. A bent slice of flat spacetime shows why the correction or a geodesic-slice prescription is necessary. For the prepared conformal scalar, changed propagation, metric variation and $\xi G\Phi^2$ supply explicit nonzero next-order source terms. The common cone, area density and consistency mechanism remain inputs. Curved stress readings are not assumed to be curved modular Hamiltonians; the common quantum source and nonlinear modular response form a strong, testable extension rather than an established consequence of the linear law.

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1 Introduction

Consciousness I asks whether a physical description can be fixed before any perspective is specified [1]. In its chiral conformal net, the geometry and modular data of a standard Cut determine each other. Transverse directions change the question. A higher-dimensional observer has a family of ball algebras, with different centres, radii and rest frames. Can one geometry account for all their readings?

For conformal matter in Minkowski spacetime these regions carry the bulk Casini–Huerta–Myers (CHM) stress charge [7]. We first construct its geometric partner rather than postulating a two-derivative conserved tensor. For every smooth perturbation the area–volume variation of a geodesic ball equals the weighted integral of the linearized Einstein tensor. The Gauss equation and intrinsic gauge invariance explain why that tensor appears. Comparing it with the CHM charge gives a conditional same-spacetime Einstein law, without a holographic dictionary. Its coefficient is supplied, and its physical consistency mechanism remains to be explained.

The next question is whether an experiment identifies the response law. A homogeneous source does not. A source with nonzero-frequency weight and an open set of boost directions identifies the entire compact intrinsic response at one tested radius, up to normalization. We retain the proof for tempered Radon Fourier measures, and provide actual scalar preparations with finite positive energy. A neutral pulse belongs to a proper stable hull; a shared non-scalar hull with translation permission can also supply a bounded positive-energy pulse. Control permission, smooth local source readings and source-apparatus costs are distinct inputs.

The physical one-sided scalar energy includes a surface charge. Keeping it in the ledger yields a raw-energy obstruction and a matched Wald coefficient; dropping it would falsely identify a geometric response. The resulting tensor is common to stress-containing selected algebras. Their relative entropies differ, not the emitted stress. A single rest frame leaves momentum and spatial stresses underdetermined, and no linear ball reading fixes free gravitational radiation.

Second order is not just a finer version of the same construction. Slice extrinsic curvature, changed source propagation, quantum reference stress and curved modular flow become relevant. We derive an extrinsic-corrected leading small-ball law and prove its two-jet equivalence to the second-order Einstein equation. A flat bent-slice counterexample and an explicit next-order scalar source test it. Section 7 states the remaining common-geometry extension with its source, readout and intervention conditions.

The linear results require a conformal net, the bulk CHM datum, and a common Minkowski cone. Q-RAIF A studies conditions for common propagation cones [5]; this paper does not derive the cone of its boosted family. Its exact conformal net is a supplied fixed-point theory. Approximate cones in a lattice realization do not by themselves establish the exact CHM modular flow used here; nonconformal corrections require a separate limit estimate [24]. The area density η and modular consistency are also inputs. Additional edge energy must be unchanged, separately read and subtracted, or directly bypassed by bulk stress readings. At second order we supply a corrected local *stress* consistency condition, not a curved CHM formula. Neither the Gauss identity nor stress conservation makes that condition physically true by itself.

Sections 2–4 establish the linear law. Sections 5–6 supply preparations, identification and partial-observer limits; Section 7 extends the geometry and source. Section 8 prices radial resolution, and the conclusion returns to the series' question of access and a common world. Indexed appendices retain the coefficient, spectral-measure, source and entropy calculations.

2 The modular family of balls

2.1 Balls, diamonds and their modular Hamiltonians

Work in Minkowski space $\mathbb{R}^{1,d}$, $d \geq 2$, with $\eta = \mathrm{diag}(-1, 1, \ldots, 1)$. A ball $B = B(x_0, R, u)$ is fixed by a centre x_0, a radius R and a future unit timelike vector u: it is the set of points x with $u \cdot (x - x_0) = 0$ and $|x - x_0| < R$, an open ball in the hyperplane $\Sigma_u(x_0)$ orthogonal to u. Its causal diamond is $D(B)$. We write

$$w_R(y) = \frac{R^2 - |y - x_0|^2}{2R}, \qquad T_{uu} = T_{\mu\nu} u^\mu u^\nu,$$

and Δ_u for the Laplacian of Σ_u.

Let $\mathcal{A}$ be a conformal net on $\mathbb{R}^{1,d}$ with vacuum Ω. The modular group of $(\mathcal{A}(D(B)), \Omega)$ acts geometrically, by the flow of the conformal Killing vector that preserves $D(B)$ [13, 14]. In coordinates with $x_0 = 0$ and $u = \partial_t$,

$$\zeta_B = \frac{1}{2R}\Big[(R^2 - t^2 - r^2)\,\partial_t - 2t\, x^i \partial_i\Big], \tag{1}$$

normalized so that its surface gravity at the edge ∂B is one; the modular period is then 2π, as for the boost of a wedge. The one-sided CHM modular energy is the stress charge

$$K_B = 2\pi \int_\Sigma T_{\mu\nu} \zeta_B^\mu n^\nu \, dV = 2\pi \int_B w_R T_{uu} \, d^d x. \tag{2}$$

Here Σ is a Cauchy surface of the diamond and the second expression uses its central ball. In a theory without an additional edge charge, a faithful type I regulator identifies it with $H_B = -\log \rho_B$, up to the additive normalization constant [7]. For a free conformal scalar the physical one-sided energy includes a surface term [8]; Section 5.4 keeps that term separate. In the continuum K_B denotes the bulk charge and its vacuum-subtracted expectations, with the required stress-smearing limits assumed finite.

The full modular generator is a different object: $\widehat{K}_B = -\log \Delta_B$. It annihilates the vacuum and implements modular covariance. At a regulated bipartition,

$$\widehat{K}_B = H_B \otimes 1 - 1 \otimes H_{B^c}^{\mathsf{T}}.$$

Thus it agrees with the physical one-sided energy in expectation differences for a unitary local to $D(B)$, whose complement state is unchanged. It agrees with the bulk charge as well when any boundary contribution is unchanged. Arbitrary state variations need not have this property. Throughout, $\delta\langle K_B \rangle$ means the vacuum-subtracted CHM charge, and modular covariance uses $\widehat{K}_B$.

2.2 Two kinds of variation

HAFF-D separates two ways in which a modular Hamiltonian can change [3], and both occur in the ball family.

A *state variation* keeps the region and changes the state: $\delta\langle K_B \rangle = 2\pi \int_B w_R \, \delta\langle T \rangle_{uu}$. For a unitary $W \in \mathcal{A}(D(B))$ with unchanged boundary charge, Araki's relative entropy of $\omega_W = \langle W\Omega, \cdot\, W\Omega \rangle$ is

$$S_B(\omega_W \| \omega) = \langle W\Omega, \widehat{K}_B W\Omega \rangle = \delta\langle K_B \rangle_{\omega_W}, \tag{3}$$

which Consciousness I uses in $1+1$ dimensions [1, 15]. For a first-order perturbation $\rho = \sigma + \epsilon\,\delta\rho$ of a regulated (type I) version, the first law is $\delta S = \delta\langle H_B\rangle$; it equals the bulk $\delta\langle K_B\rangle$ when the boundary variation is zero.

A *region variation* keeps the state and moves the ball. The family of all diamonds is one orbit of the conformal group, of dimension $2(d+1)$: $d+1$ for the centre, one for the radius and d for the rest frame. A one-parameter conformal group e^{-iaP_ξ} moves $D(B)$ to $D(B_a)$ and $\widehat{K}_{B_a} = e^{-iaP_\xi}\widehat{K}_B e^{iaP_\xi}$, so

$$\delta\widehat{K}_B = -i\,\delta a\,[P_\xi, \widehat{K}_B], \qquad \langle\Omega, \delta\widehat{K}_B\,\Omega\rangle = 0, \tag{4}$$

because $P_\xi\Omega = 0$ and $\widehat{K}_B\Omega = 0$. This is the variation formula of HAFF-D for a half-sided modular family. Null translations of a wedge give half-sided modular inclusions [16, 17], and balls are conformal images of wedges, so the supplied ball family has the region interface required by HAFF-D's conditional holographic matching. Vacuum covariance concerns the full generator.

This transitivity concerns a single diamond, not an arbitrary pair. A nested pair of balls carries the inversive-distance modulus; it is not determined by a dimension count for the orbit of one ball.

2.3 Conservation from the diamond

The algebra of the diamond is the same for every Cauchy surface of $D(B)$. For its bulk stress charge, independence of the first form of (2) from Σ is exactly conservation and tracelessness.

Proposition 2.1 (Conservation from the diamond)**.** *Let $T_{\mu\nu}$ be a smooth symmetric tensor field on an open set $U \subset \mathbb{R}^{1,d}$. The charge $Q_\Sigma(B) = \int_\Sigma T_{\mu\nu}\zeta_B^\mu n^\nu dV$ is the same for all Cauchy surfaces Σ of $D(B)$, for every diamond with $\overline{D(B)} \subset U$, if and only if $\partial^\mu T_{\mu\nu} = 0$ and $T^\mu{}_\mu = 0$ in U.*

Proof. ζ_B is a conformal Killing vector, $\partial_\mu\zeta_\nu + \partial_\nu\zeta_\mu = \frac{2}{d+1}(\partial\cdot\zeta_B)\eta_{\mu\nu}$, with $\partial\cdot\zeta_B = -(d+1)t/R$ in the coordinates of (1). Hence

$$\partial^\mu\big(T_{\mu\nu}\zeta_B^\nu\big) = \zeta_B^\nu\,\partial^\mu T_{\mu\nu} + \frac{\partial\cdot\zeta_B}{d+1}\,T^\mu{}_\mu.$$

Two Cauchy surfaces of $D(B)$ share the edge ∂B and bound a region of $D(B)$; if the right side vanishes, Stokes' theorem gives equal charges. Conversely, deforming Σ near an interior point x shows that the left side vanishes at every $x \in D(B)$, for every diamond. As B varies over diamonds containing a fixed x, the pairs $(\zeta_B(x), \partial\cdot\zeta_B(x))$ span $\mathbb{R}^{d+1}\times\mathbb{R}$, so both terms vanish separately. □

For the anomaly-free flat conformal stress both properties hold. The proposition identifies what slice-independent stress-charge representation requires: a modular datum is attached to a region's algebra, not to a slice; provided this same bulk stress charge represents it on every Cauchy surface, the stated equality forces conservation and a vanishing trace. Conservation is therefore not an observation to be placed beside the Bianchi identity; it is a property of the modular family.

The finite-radius charge has a controlled expansion with leading coefficient $c_0 = V_d/(d+2)$; its exact Fourier transfer and all radial coefficients are given in Appendix A.1. In particular, an open set of normals recovers a smooth source tensor from its leading small-ball charges (Corollary A.2).

3 The geometric functional of a ball

3.1 What a ball functional may read

Let $h_{\mu\nu}$ be a smooth metric perturbation of Minkowski space, and h_{ij} the perturbation of the metric induced on $\Sigma_u(x_0)$.

Definition 3.1 (Ball functional)**.** A *ball functional* assigns to every ball $B = B(x_0, R, u)$ and every h a number $\delta F_B[h]$ such that:

(F1) it is linear in h;

(F2) it depends only on the induced perturbation h_{ij} on $\overline{B}$;

(F3) it is unchanged by $h_{ij} \mapsto h_{ij} + \partial_i\xi_j + \partial_j\xi_i$ for every smooth vector field ξ on a neighbourhood of $\overline{B}$ in Σ_u;

(F4) it is invariant under the orthogonal group of Σ_u fixing x_0;

(F5) it has the same form for every centre and every rest frame, its dependence on R and on any constants being arbitrary;

(F6) it is a distribution of finite order supported in $\overline{B}$.

No scale covariance is assumed, so the functional may contain any number of length scales besides R. The restrictive condition is (F2): the functional reads the intrinsic geometry of the ball. Three considerations support it. The region to which the modular datum belongs is the ball on its slice, whose geometric data are the induced metric and the extrinsic curvature. The extrinsic curvature reverses sign under reflection of time about Σ_u, while $\delta\langle K_B\rangle$ does not change when the state is reflected in a theory with that symmetry; a linear functional matched to the modular energy for all states cannot depend on it. And second time derivatives of h are not data of the region at all: reading them presupposes an evolution law for the metric, which is what one wants to obtain. Proposition 4.7 shows what happens when (F2) is dropped.

Lemma 3.2 (Ball functionals see only the Einstein tensor)**.** *Let δF satisfy (F1)–(F4) and (F6), and let $\tau_u = \partial_i\partial_j h_{ij} - \Delta_u h_{kk}$ be the linearized scalar curvature of Σ_u.*

(a) For a plane wave $h_{ij} = \mathrm{Re}(\varepsilon_{ij}e^{ip\cdot x})$ on Σ_u, $\delta F_B[h] = \mathrm{Re}\big(m_B(|p|)\,\hat{\tau}\,e^{ip\cdot x_0}\big)$ with $\hat{\tau} = p^2\varepsilon_{kk} - p_ip_j\varepsilon_{ij}$ and m_B an even entire function, the transfer function *of the functional.*

(b) For polynomial h, $\delta F_B[h] = \sum_k m_k\,\Delta_u^k\tau_u(x_0)$, where $m_B(q) = \sum_k m_k(-q^2)^k$.

(c) $\tau_u = 2\,G^{\mathrm{lin}}_{uu}$.

(d) There is a rotation-invariant scalar distribution μ_B, supported in $\overline{B}$, such that for every smooth h on a neighbourhood of that ball,

$$\delta F_B[h] = \langle \mu_B, \tau_u[h]\rangle, \qquad \widehat{\mu}_B(p) = m_B(|p|).$$

No growth condition on h outside the ball is required.

The proof is in Appendix A.3.

At linear order about a flat slice, the Gauss equation equates half the intrinsic scalar curvature with the normal–normal Einstein component. The quadratic extrinsic term is essential beyond that order (Section 7). The Bianchi identity, which makes G^{lin} divergence-free, therefore comes with the functional; it is not a separate premise.

Corollary 3.3 (The Weyl curvature is invisible)**.** *If h is smooth and $G^{\mathrm{lin}} = 0$, every ball functional vanishes on h for every ball. In particular, for $d \geq 3$ a free gravitational wave, whose Riemann tensor is pure Weyl curvature, gives zero for every ball, every rest frame and every functional satisfying (F1)–(F4) and (F6).*

3.2 The functional

For a ball $B = B(x_0, R, u)$ and a metric $g = \eta + \epsilon h$, let B_ϵ be the geodesic ball of radius R about x_0 in the metric induced on $\Sigma_u(x_0)$, A_ϵ the area of its boundary and V_ϵ its volume. Define

$$F_B[g] = A(\partial B_\epsilon) - \frac{d-1}{R}\, V(B_\epsilon), \qquad \delta F_B[h] = \frac{d}{d\epsilon} F_B[\eta + \epsilon h]\Big|_{\epsilon=0}. \tag{5}$$

For $d = 2$ the area is the length of the boundary circle.

Theorem 3.4 (Exact identity)**.** *For every smooth h and every ball,*

$$\delta F_B[h] = -\frac{1}{2}\int_B w_R\, \tau_u\, d^d x = -\int_B w_R\, G^{\mathrm{lin}}_{uu}\, d^d x. \tag{6}$$

Proof. Put $x_0 = 0$ and write points of Σ_u as $s\hat{n}$. To first order the geodesic distance from 0 to $s\hat{n}$ is $s + \frac{1}{2}\int_0^s h_{nn}(\sigma\hat{n})\, d\sigma$, because the length functional is stationary on the straight line. The boundary of B_ϵ is therefore $s = R - \epsilon\Phi(\hat{n})$ with $\Phi(\hat{n}) = \frac{1}{2}\int_0^R h_{nn}(\sigma\hat{n})\, d\sigma$. The area element of a surface changes by half the trace of h over its tangent space, and the volume element by half the full trace, so

$$\delta A = \frac{1}{2}\oint_{S_R} \big(h_{kk} - h_{nn}\big)\, dA - (d-1)R^{d-2}\oint \Phi\, d\Omega, \qquad \delta V = \frac{1}{2}\int_B h_{kk}\, d^d x - R^{d-1}\oint \Phi\, d\Omega.$$

In $\delta A - \frac{d-1}{R}\delta V$ the terms in Φ cancel:

$$\delta F_B = \frac{1}{2}\oint_{S_R} \big(h_{kk} - h_{nn}\big)\, dA - \frac{d-1}{2R}\int_B h_{kk}\, d^d x.$$

On the other side use $w_R = 0$ and $\partial_n w_R = -1$ on S_R, $\partial_i w_R = -x_i/R$ and $\partial_i\partial_j w_R = -\delta_{ij}/R$. Integrating by parts twice, $\int_B w_R\, \partial_i\partial_j h_{ij} = \oint h_{nn}\, dA - \frac{1}{R}\int_B h_{kk}$ and $\int_B w_R\, \Delta h_{kk} = \oint h_{kk}\, dA - \frac{d}{R}\int_B h_{kk}$. Subtracting gives $\int_B w_R \tau_u = \oint (h_{nn} - h_{kk})\, dA + \frac{d-1}{R}\int_B h_{kk}$, which is $-2\,\delta F_B$. The second equality in (6) is Lemma 3.2(c). □

The geometric identity can also be evaluated without the linearized formulas, by shooting geodesics with their Jacobi fields in the metric $\delta + \epsilon h$ and differencing in ϵ. For a non-conformally-flat perturbation, this evaluation agrees with the right side of (6) to relative accuracy 2×10^{-11} in $d = 2$ and 10^{-10} in $d = 3$.

The content of Theorem 3.4 is known: it is the flat-space, linear geometric part of the first law of causal diamonds of Jacobson and Visser [21], there obtained from the covariant phase space of general relativity with the Einstein constraint imposed. The proof above uses no field equation. That is what the next section needs: (6) holds for every perturbation, whatever its dynamics.

3.3 Where the coefficient comes from

The volume term in (5) is the fixed-volume prescription of Jacobson's entanglement-equilibrium argument [19], where it is motivated physically. In the modular family it has a structural origin.

Proposition 3.5 (The coefficient)**.** *Let $F_\lambda = A - \lambda V$ for geodesic balls.*

(a) Fix a reference radius R_0 and hold λ_0 fixed when varying R. Then

$$\frac{d}{dR}\left[\Omega_{d-1}\left(R^{d-1} - \frac{\lambda_0 R^d}{d}\right)\right]\Bigg|_{R=R_0} = 0 \quad \Longleftrightarrow \quad \lambda_0 = \frac{d-1}{R_0}.$$

Centre and frame moves leave the flat value unchanged for every λ_0. This fixes the Lagrange multiplier for the variation at R_0.

(b) Write $\delta F_\lambda = \sum_k \gamma_k(\lambda)\, R^{d+1+2k} \Delta_u^k \tau_u(x_0)$ for polynomial h. Then

$$\frac{\gamma_k(\lambda)}{-c_k/2} = \frac{d+2k+2-\lambda R}{2k+3},$$

which is independent of k if and only if $\lambda = (d-1)/R$. Thus δF_λ has the modular weight w_R at every order in R exactly for this coefficient.

The proof is in Appendix A.4.

Agreement of a geometric functional with the modular data at leading order in R does not fix λ; only the coefficient of the Einstein equation would change. Matching the full kernel fixes it to the fixed-reference multiplier. The family $F_R = A_R - (d-1)V_R/R$ itself is not stationary: $F_R[\eta] = \Omega_{d-1} R^{d-1}/d$, so its radius derivative is $\Omega_{d-1}(d-1)R^{d-2}/d$. These two variations must be distinguished.

4 Modular consistency and the linearized Einstein equation

Definition 4.1 (Modular consistency)**.** Let a state have smooth stress-tensor expectation $\delta\langle T\rangle$, let h be a metric perturbation, δF a ball functional and $\eta > 0$ a constant. The pair satisfies (MC_η) on a family of balls if

$$\eta\, \delta F_B[h] + \delta\langle K_B\rangle = 0 \tag{7}$$

for every ball of the family.

Here K_B is the bulk datum defined by (2). If the physical one-sided modular energy is $H_B = K_B + Q_B$, the same condition uses $\delta\langle H_B\rangle - \delta\langle Q_B\rangle$. It uses the raw physical energy only when the boundary variation vanishes.

The constant η has the dimension of an inverse area. For first-order perturbations with unchanged boundary charge, where $\delta S_B = \delta\langle K_B\rangle$, and with $\delta F = \delta F_B$ of (5), condition (7) says that $\eta A + S$ is stationary at fixed volume, which is Jacobson's entanglement equilibrium [19]. We keep the modular energy, not the entropy, in (7); Remark 4.8 and Section 6 explain why. In words, (MC_η) says that the modular energy of every ball is balanced by one geometric response of the ball, with one constant. If a boundary charge varies, the first-order entropic version uses $\delta S_B - \delta\langle Q_B\rangle$, not the uncorrected entropy variation.

4.1 Equivalence

Theorem 4.2 (Equivalence)**.** *Let $U \subset \mathbb{R}^{1,d}$ be open, and let $\delta\langle T\rangle$ and h be smooth on U. With δF the functional* (5)*, the following are equivalent.*

(a) (MC_η) *holds for all balls $\overline{B} \subset U$, in all rest frames and at all radii.*

(b) (MC_η) *holds to leading order in R at every $x_0 \in U$, for all u in some nonempty open set of rest frames.*

(c) $G^{\mathrm{lin}}_{\mu\nu} = \dfrac{2\pi}{\eta}\,\delta\langle T\rangle_{\mu\nu}$ *in U.*

Proof. (c)⇒(a): by Theorem 3.4, $\eta\,\delta F_B = -\eta\int_B w_R G^{\mathrm{lin}}_{uu} = -2\pi\int_B w_R\delta\langle T\rangle_{uu} = -\delta\langle K_B\rangle$. (a) ⇒(b) is immediate. (b)⇒(c): the leading terms of both sides of (7) carry the same factor $V_d R^{d+1}/(d+2)$ (Proposition A.1 and Theorem 3.4), so $\eta G^{\mathrm{lin}}_{uu}(x_0) = 2\pi\delta\langle T\rangle_{uu}(x_0)$ for an open set of u, and Corollary A.2 gives the tensor equation. □

Normalization. The factor $V_d/(d+2)$ appears on both sides and cancels, so the coupling is $2\pi/\eta$ in every dimension. With $\eta = 1/4G$ it is $8\pi G$: the 2π is the modular period of a ball and the $1/4$ is the Bekenstein–Hawking area density.

All radii from leading order. Agreement at leading order already forces the field equation, and the field equation makes the two sides agree at every radius. The finite-size corrections of the modular side and of the geometric side are therefore identical, term by term in (24); for a Fourier mode both have the transfer function f of (25). Given the field equation, the full ball-dependent functional agrees with the modular data on every ball; this is the converse direction.

Conservation and trace. The Einstein tensor is divergence-free by the Bianchi identity and $\delta\langle T\rangle$ is conserved and traceless by Proposition 2.1; the equation is consistent with both and requires $R^{\mathrm{lin}} = 0$ in U. Neither property is an additional premise.

Background. The background is Minkowski space, so no cosmological term appears. For diamonds in de Sitter or anti-de Sitter space the first law of Jacobson and Visser contains the cosmological constant and its variation [21]; the corresponding version of Theorem 3.4 is not derived here.

4.2 Uniqueness

The canonical identity holds for every metric perturbation. Identifying that response from data requires a source that excites more than the zero-frequency channel.

Lemma 4.3 (Conserved amplitudes under frame variation)**.** *Let $k \neq 0$, and let t, g be symmetric complex tensors with $k^\mu t_{\mu\nu} = k^\mu g_{\mu\nu} = 0$, $t \neq 0$. On a nonempty open set of future unit timelike vectors suppose $g_{uu} = a(q_u)t_{uu}$, where $q_u^2 = k^2 + (u\cdot k)^2$ and a is analytic on an interval of attained momenta. Then $g = a_0 t$ for a constant a_0. Tracelessness is not needed.*

Proof. For timelike k, choose its rest frame. Conservation sets $g_{0a} = t_{0a} = 0$. Write $u = (\sqrt{1+r^2}, rn)$. At fixed $r > 0$, equality on an open patch of the sphere implies $g_{ij} = a(q_r)t_{ij}$ by equality of quadratic forms. The nonzero tensor t fixes the same scalar as r varies.

For spacelike k, put it along the first spatial coordinate. Conservation sets $g_{1a} = t_{1a} = 0$. At fixed $u^1 = b$, the remaining coordinates run over the timelike hyperboloid of squared norm $-1-b^2$. A homogeneous quadratic form vanishing on an open patch of this hyperboloid vanishes on the open timelike cone by analytic continuation and homogeneity, hence identically. Thus $g = a(q_b)t$, with one scalar independent of b.

For null k, take $k \propto (1, 1, 0, \ldots)$. Conservation gives $g_{0a} + g_{1a} = t_{0a} + t_{1a} = 0$. At fixed $r = u^0 - u^1 > 0$, the transverse coordinates v vary freely, q_u is fixed, and

$$g_{uu} = g_{00} r^2 + 2 r g_{0A} v^A + g_{AB} v^A v^B .$$

Equality on an open patch is equality of these polynomials. It gives $g = a(q_r)t$; again $t \neq 0$ fixes the scalar. An open frame set contains patches with an interval of r or b, away from any isolated critical value of q_u. □

Theorem 4.4 (Nonzero-frequency identification)**.** *Fix $R > 0$ and let δF obey (F1)–(F6), with transfer function m_R. Suppose*

$$\delta\langle T\rangle_{ab}(x) = \mathrm{Re}(t_{ab} e^{ik\cdot x}), \qquad G^{\mathrm{lin}}_{ab}[h](x) = \mathrm{Re}(g_{ab} e^{ik\cdot x}),$$

where $k \neq 0$, $t \neq 0$, and t, g are conserved symmetric amplitudes. If (MC_η) holds on a nonempty open set of centres and on a nonempty open set of rest frames at this radius, then

$$m_R(q) = -\frac{c}{2}\widehat{w}_R(q) \quad \textit{for every } q, \qquad g_{ab} = \frac{2\pi}{c\eta} t_{ab}, \qquad c \in \mathbb{R} \setminus \{0\}.$$

Consequently $\delta F = c\, \delta F_B$ at this radius on arbitrary smooth metric inputs. If the same witness satisfies consistency at further radii, the same c applies at each tested radius. A hypothesis restricted to $R < R_0$ identifies only that range.

More generally let the spacetime Fourier transforms of $\delta\langle T\rangle$ and $G^{\mathrm{lin}}[h]$ be tempered tensor-valued complex Radon measures, meaning that their total variations are locally finite and have at most polynomial growth, with the source measure nonzero away from $k = 0$. With consistency at all spacetime centres and an open frame set, the response is identified at this radius and $G^{\mathrm{lin}}[h] = (2\pi/(c\eta))\delta\langle T\rangle$ as distributions. This includes continuous Fourier densities and sources stationary in one frame with a continuous spatial spectrum. General tempered distributions need a separate argument.

The proof is in Appendix A.5.

One radius is enough because changing the rest frame turns a nonzero spacetime frequency into an interval of spatial frequencies, and the compact support of the response supplies analytic continuation. Exact identification does not imply stable identification from finitely many noisy frames. The theorem concerns response laws tested by such data, rather than a physical derivation of the modular-consistency hypothesis.

Corollary 4.5 (The spectral criterion is necessary and sufficient)**.** *Let $d \geq 2$, fix a radius, and suppose a nonzero smooth source and a smooth metric satisfy (MC_η) for some functional obeying (F1)–(F6), at every centre and on an open set of frames. Suppose both tensor Fourier transforms belong to the tempered Radon class of Theorem 4.4. The data identify the canonical functional up to normalization if and only if the source measure has nonzero weight away from zero. If it is supported at zero, both tensors are constant and the data test only $m_R(0)$.*

The proof is in Appendix A.6.

The nonzero-source hypothesis excludes a degenerate experiment: if both source and Einstein tensor vanish, every admissible functional fits. Polynomial sources fall outside the Radon class; their Fourier transforms contain derivatives of δ_0. A polynomial tensor of degree n probes only the finitely many coefficients m_k with $2k \leq n$, so it also leaves higher spatial derivatives undetermined.

Example 4.6 (A homogeneous source leaves the response open)**.** In $D = d + 1 \geq 3$ spacetime dimensions put $T_{ab} = \mathrm{diag}(d, 1, \ldots, 1)$, $\kappa = 2\pi/\eta$, $x^2 = \eta_{ab}x^a x^b$, and $Txx = T_{ab}x^a x^b$. The quadratic metric

$$h_{ab} = -\frac{\kappa}{2(D-2)}\big(T_{ab}x^2 + \eta_{ab}Txx\big)$$

has $R^{\mathrm{lin}} = 0$ and $G^{\mathrm{lin}}_{ab} = \kappa T_{ab}$. For every $\ell \neq 0$, the intrinsic functional

$$\delta F_B^{\mathrm{new}}[h] = \delta F_B[h] + \ell^2 \int_B w_R \Delta_u \tau_u[h]\, d^d x$$

obeys (F1)–(F6) and satisfies exactly the same consistency condition on this pair at every centre, radius and frame. Its extra term vanishes because G^{lin} is constant. It is not a multiple of the canonical functional: its transfer function is $(-\frac{1}{2} - \ell^2 q^2)\widehat{w}_R(q)$. A homogeneous source tests only the zero-momentum response.

The Einstein tensor appears through intrinsic slice curvature and the Gauss equation. Nonzero-frequency data then identify its two-derivative response: no independent assumption excluding internal length scales is needed. Actual source preparation is therefore part of what makes a geometric law identifiable.

4.3 What fails without each hypothesis

Proposition 4.7. *(a)* One order in R is not enough. *The area of the geodesic sphere at fixed radius, $\lambda = 0$ in Proposition 3.5, satisfies* (MC_η) *to leading order with $G^{\mathrm{lin}} = \frac{6\pi}{(d+2)\eta}\delta\langle T\rangle$, but its coefficient of order R^{d+3} differs from the modular one by the factor $1 - \frac{2(d-1)}{5(d+2)}$ (-16% for $d = 3$). Consistency at that order would require $\Delta_u \delta\langle T\rangle_{uu} = 0$ in every frame. For a Fourier mode the two transfer functions differ by 0.29%, 1.2%, 5.2% and 14% at $R|k_u| = 0.5, 1, 2, 3$ ($d = 3$).*

(b) Reading off the slice. *The functional $\delta F_B^{(\ell)} = -\int_B w_R\,(G^{\mathrm{lin}}_{uu} - \ell^2 \Box G^{\mathrm{lin}}_{uu})$ satisfies (F1) and (F3)–(F6) but not (F2).* (MC_η) *holds for it at all radii and in all frames if and only if $(1 - \ell^2\Box)G^{\mathrm{lin}}_{\mu\nu} = (2\pi/\eta)\delta\langle T\rangle_{\mu\nu}$, a fourth-order equation.*

(c) Reading only the trace-free part. *The functional $-\int_B w_R\,(R^{\mathrm{lin}}_{uu} + R^{\mathrm{lin}}/(d+1))$ gives $R^{\mathrm{lin}}_{\mu\nu} - \frac{1}{d+1}\eta_{\mu\nu}R^{\mathrm{lin}} = (2\pi/\eta)\delta\langle T\rangle_{\mu\nu}$; the Bianchi identity makes R^{lin} constant, and the result is $G^{\mathrm{lin}}_{\mu\nu} + \Lambda\eta_{\mu\nu} = (2\pi/\eta)\delta\langle T\rangle_{\mu\nu}$ with an integration constant $\Lambda = \frac{d-1}{2(d+1)}R^{\mathrm{lin}}$.*

Proof. (a) From Proposition 3.5(b) with $\lambda = 0$ the ratios to the modular coefficients are $(d + 2)/3$ and $(d+4)/5$ at the first two orders; the transfer functions are computed from the linearized geodesic ball. (b) and (c) follow as in Theorem 4.2, with the operator inside the integral in place of G^{lin}_{uu}; in (c), $\partial^\mu R^{\mathrm{lin}}_{\mu\nu} = \frac{1}{2}\partial_\nu R^{\mathrm{lin}}$ and $\partial^\mu \delta\langle T\rangle_{\mu\nu} = 0$ give $\big(\frac{1}{2} - \frac{1}{d+1}\big)\partial_\nu R^{\mathrm{lin}} = 0$. □

Part (b) is the general lesson. Higher-derivative laws arise when the geometric response of a region depends on how the geometry evolves away from the region's slice; the analysis of higher-curvature entanglement equilibrium by Bueno et al. fits this pattern [25]. Part (c) shows where a cosmological constant could enter as an integration constant: only through a functional blind to the trace, which the intrinsic one is not.

Remark 4.8 (Energy, not entropy). For a unitary $W \in \mathcal{A}(D(B))$ with unchanged boundary charge, the entropy of the ball does not change at a regulator preserving that local unitary, while $\delta\langle K_B\rangle = S_B(\omega_W\|\omega) \geq 0$, with strict inequality whenever the local state changes, by (3).

For a coherent excitation strictly local to $D(B)$, the modular energy and the relative entropy of that ball are equal and of second order in the amplitude. A condition written with entropy, $\eta\,\delta F_B + \delta S_B = 0$, would then require $\delta F_B = 0$ on that ball, in conflict with the semiclassical Einstein response to a nontrivial preparation. Condition (7) has no such problem. Entanglement equilibrium agrees with it for perturbations whose modular energy is of first order, and only for those.

5 What an observer's ball data fix

An observer does not have all balls. It has balls near its worldline, of sizes between a resolution and a scale of interest, in the rest frames it can compare. This section asks what such data fix through (MC_η), and with what stability.

5.1 One rest frame

Proposition 5.1 (One rest frame gives the Hamiltonian constraint)**.** *Fix u.*

(a) (MC_η) *for all balls in the frame u, at all centres in U and all radii, is equivalent to $G^{\mathrm{lin}}_{uu} = (2\pi/\eta)\delta\langle T\rangle_{uu}$ in U.*

(b) Let $\delta\langle T\rangle$ be stationary in the frame u, and let $\delta\langle T\rangle'$ be its time reverse, $T'_{00} = T_{00}$, $T'_{0i} = -T_{0i}$, $T'_{ij} = T_{ij}$. Then $\delta\langle T\rangle'$ is conserved, traceless and stationary, and the modular energies of all balls at rest in u are the same for both sources. For a ball boosted with rapidity β in the direction n they differ by $2\pi\int_B w_R\, 2\sinh(2\beta)\, n^i T_{0i}$.

(c) Consequently, if h satisfies $G^{\mathrm{lin}} = (2\pi/\eta)\delta\langle T\rangle$, the pair $(\delta\langle T\rangle', h)$ satisfies (MC_η) *for every ball at rest in u, although $G^{\mathrm{lin}}_{0i} = -(2\pi/\eta)T'_{0i}$.*

Proof. (a) is Theorem 4.2 with a single frame, where the leading term gives only the uu component. (b) Stationarity and conservation give $\partial_i T_{0i} = -\partial_t T_{00} = 0$ and $\partial_j T_{ij} = -\partial_t T_{0i} = 0$, which are the conservation equations of $\delta\langle T\rangle'$; the trace is unchanged. Balls at rest in u read only T_{00}. For $u = (\cosh\beta, \sinh\beta\, n)$, $T_{uu} - T'_{uu} = 4\cosh\beta\sinh\beta\, n^i T_{0i}$. (c) follows from (a) and (b). □

A body rotating one way and the same body rotating the other way have identical ball data in their common rest frame, and opposite frame dragging. The momentum constraint and the spatial components of the field equation, which carry the dynamics, are the part of the geometry that only comparison of rest frames reveals. In this precise sense the multi-directional character of the data is where the gravitational dynamics sits.

5.2 Several rest frames and their stability

By Corollary A.2 any open set of rest frames suffices in exact arithmetic. With noise and finite balls the question is quantitative.

Proposition 5.2 (Rapidity sets the precision)**.** *Let the observer use its rest frame and boosts of rapidity β in a fixed set of directions for which the leading-order readings determine a traceless symmetric tensor, and let each normalized reading $y_a = \delta\langle T\rangle_{u_a u_a}(x_0) + e_a$ carry an independent error of variance σ^2. As $\beta \to 0$ the least-squares estimate has root-mean-square error of order σ in T_{00}, σ/β in T_{0i} and σ/β^2 in T_{ij}.*

Proof. With $u_a = (1+\frac{\beta^2}{2}, \beta n_a)+O(\beta^3)$, $y_a = T_{00}+2\beta\, n_a^i T_{0i}+\beta^2(T_{00}+n_a^i n_a^j T_{ij})+O(\beta^3)+e_a$. The columns of the design matrix for (T_{00}, T_{0i}, T_{ij}) scale as $1, \beta, \beta^2$, and for a direction set whose leading blocks are nondegenerate the inverse scales as $1, \beta^{-1}, \beta^{-2}$. □

Finite balls add a bias. A reading at radius R multiplies each Fourier component by $f(R|k_u|)$ (Proposition A.1), and $|k_u|$ grows like e^β for large boosts. For the stresses the bias then grows like $R^2 e^{2\beta}|k|^2$ while the noise falls like $\sigma e^{-2\beta}$, and the best error obtainable with one radius scales as $\sqrt{\sigma}\, R\, |k|$. Table 1 shows both regimes for one mode in $d = 3$, with thirteen frames (the rest frame and boosts along the six coordinate half-axes and six face diagonals). A finer scan gives an optimal error proportional to $R^{1.03}$ when noise dominates and to $\sigma^{0.48}$ between $\sigma = 10^{-4}$ and 10^{-2} at $R = 0.02L$; at very small σ the rest-frame bias, proportional to R^2, dominates. Since the transfer function is known exactly, readings at several radii can remove the finite-size bias. What cannot be removed is the amplification of noise by a small range of rapidities.

	$\sigma = 10^{-4}$			$\sigma = 10^{-2}$		
β	T_{00}	T_{0i}	T_{ij}	T_{00}	T_{0i}	T_{ij}
0.05	3.2×10^{-5}	1.1×10^{-3}	8.8×10^{-2}	3.3×10^{-3}	1.0×10^{-1}	9.3
0.2	3.1×10^{-5}	2.7×10^{-4}	5.8×10^{-3}	3.1×10^{-3}	2.7×10^{-2}	5.8×10^{-1}
0.7	3.0×10^{-5}	7.6×10^{-5}	4.3×10^{-4}	1.9×10^{-3}	5.3×10^{-3}	3.8×10^{-2}
1.0	5.3×10^{-5}	9.5×10^{-5}	3.4×10^{-4}	1.1×10^{-3}	3.1×10^{-3}	1.7×10^{-2}
2.0	4.5×10^{-4}	7.3×10^{-4}	2.2×10^{-3}	4.6×10^{-4}	8.2×10^{-4}	2.7×10^{-3}

Table 1: Root-mean-square reconstruction error of a unit-norm traceless conserved source $\mathrm{Re}(\varepsilon_{\mu\nu} e^{ik\cdot x})$, $k = (1, 0.6, 0.3, -0.2)/L$, from leading-order readings of balls of radius $R = 0.05L$ in thirteen frames of rapidity β, 200 noise draws.

5.3 Free radiation

By Corollary 3.3, (MC_η) fixes h only up to solutions of $G^{\mathrm{lin}} = 0$. For $d \geq 3$ these include gauge modes and free gravitational waves. This is exactly the freedom that the Einstein equation itself leaves. No ball functional registers a free wave at linear order, in any frame and at any radius. At second order its quadratic geometry enters $G^{(2)}[h, h]$. Moving that term to an effective source is an additional convention, not an extra matter stress tensor. Shape-sensitive regions can also probe tidal data: a deformed region couples to the tidal part of the curvature. In $2+1$ dimensions the Riemann tensor is algebraically determined by the Ricci tensor, so $G^{\mathrm{lin}} = 0$ implies a locally pure-gauge perturbation on a contractible domain. Global identifications and boundary conditions can still distinguish metrics.

5.4 A locally prepared source and its boundary datum

The source witness of Theorem 4.4 can be prepared by a local operation. For a free scalar one must also specify which part of its modular energy is being read. We use the scalar boundary prescription of Casini, Mazzitelli and Testé [8, Eq. (58)], with the boundary-condition derivation of Herzog and Nishioka [11, Secs. 1–2]. The prescription agrees with thermal numerical calculations and Rényi-entropy checks [9, 10]. For the conformal improved stress tensor it gives

$$H_B = K_B + Q_B, \qquad Q_B = 2\pi\xi \int_{\partial B} \sum_a :\phi_a^2:\, dA, \qquad \xi = \frac{d-1}{4d}. \tag{8}$$

There is one real component for a real scalar and two for a complex scalar. The formula uses vacuum-subtracted expectations. A surface expectation is defined as the limit of smooth collar smearings of the normal-ordered field square, whose weights converge to surface measure. No sharp boundary operator on arbitrary states is assumed.

Proposition 5.3 (Local preparation supplies a nonzero spectral witness)**.** *In the free massless complex scalar vacuum, let f be a real smooth compactly supported spacetime function with nonzero one-particle class. Put $W_\theta = \exp[i\theta\phi_1(f)]$, $\theta \neq 0$, and $\varphi_f = \Delta_{\mathrm{PJ}} f$, with the sign convention fixed so that $\omega_{W_\theta}(\phi_1(x)) = \theta\varphi_f(x)$. Then*

$$\delta\langle T\rangle_{\mu\nu} = \theta^2 \left[\partial_\mu\varphi_f\partial_\nu\varphi_f - \frac{1}{2}\eta_{\mu\nu}(\partial\varphi_f)^2 + \xi(\eta_{\mu\nu}\Box - \partial_\mu\partial_\nu)\varphi_f^2\right]. \tag{9}$$

It is smooth, conserved and traceless, has finite strictly positive total spatial energy, and has a finite tensor-valued Radon Fourier measure with nonzero mass away from $k = 0$. All polynomial moments of that measure are finite. Its boundary-energy variation is

$$\delta\langle Q_B\rangle = 2\pi\xi\theta^2 \int_{\partial B} \varphi_f^2 \, dA. \tag{10}$$

If supp *f is compactly contained in $D(B)$, this term vanishes for that ball. It need not vanish for the same state on other balls.*

The proof is in Appendix B.1.

Example 5.4 (A common state, source and bulk response)**.** Take the prepared state above and $\kappa = 2\pi/\eta$. A smooth linearized response $G^{\mathrm{lin}}[h] = \kappa\delta\langle T\rangle$ exists with constraint-compatible initial data. For example, write $\bar{h}_{\mu\nu} = h_{\mu\nu} - \eta_{\mu\nu}h^\alpha{}_\alpha/2$ and solve $\Box\bar{h}_{\mu\nu} = -2\kappa\delta\langle T\rangle_{\mu\nu}$. At a fixed time t_0, choose smooth Poisson solutions

$$\Delta a = -2\kappa\delta\langle T\rangle_{00}(t_0,\cdot), \qquad \Delta b_j = 2\kappa\delta\langle T\rangle_{0j}(t_0,\cdot).$$

Set $\bar{h}_{00} = a$, all other initial components zero, $\partial_t\bar{h}_{0\nu} = 0$, and $\partial_t\bar{h}_{ij} = \partial_i b_j + \partial_j b_i - \delta_{ij}\partial_\ell b_\ell$. The Lorenz constraint $C_\nu = \partial^\mu\bar{h}_{\mu\nu}$ and its first time derivative vanish initially. Conservation gives $\Box C_\nu = 0$, so they vanish throughout the linear evolution and $G^{\mathrm{lin}}[h] = \kappa\delta\langle T\rangle$. Compact smooth sources on a slice admit these Poisson solutions; in $d = 2$ logarithmic behaviour at spatial infinity is allowed. The construction is smooth and tempered, and is perturbatively small on each fixed compact domain as $\theta \to 0$.

For this pair Theorem 4.2 gives, for every ball,

$$\eta\,\delta F_B[h] + \delta\langle H_B\rangle - \delta\langle Q_B\rangle = 0. \tag{11}$$

Both $\widehat{\delta\langle T\rangle}$ and $\widehat{G^{\mathrm{lin}}[h]} = \kappa\widehat{\delta\langle T\rangle}$ meet the Radon hypotheses. Hence one radius, all centres and an open frame family identify the canonical intrinsic response from these edge-subtracted readings. The matter state, its physical source and the response law coexist here; the Einstein coupling and the constraint-compatible gravitational data were supplied, not generated by the Weyl preparation.

The subtraction has operational content. It requires access to calibrated collar readings of $\sum_a :\phi_a^2:$, in addition to the modular energy or the bulk stress readings. Their limiting expectations are finite in this coherent family, but no finite sample, resolution or preparation cost for the collar limit has been derived. The total-energy formula counts the prepared radiation, not the switching, phase-reference or reset costs of the source apparatus. If the bulk stress charge is read directly, the subtraction is unnecessary; calling that charge the raw physical H_B would still be incorrect.

Corollary 5.5 (Raw scalar energy cannot obey the bulk law on all small balls)**.** *At every centre x_0 with $\varphi_f(x_0) \neq 0$, and in any one fixed rest frame, no smooth metric perturbation satisfies $\eta\,\delta F_B[h] + \delta\langle H_B\rangle = 0$ for all sufficiently small radii. Thus the obstruction is visible at each point of a nonempty open set for every nonzero coherent preparation.*

Proof. At a point with $\varphi_f(x_0) \neq 0$,

$$\delta\langle Q_B\rangle = 2\pi\xi\theta^2\Omega_{d-1}R^{d-1}\varphi_f(x_0)^2 + O(R^{d+1}).$$

The bulk charge and the canonical intrinsic geometric variation are both $O(R^{d+1})$ for smooth source and metric fields. They cannot cancel the nonzero R^{d-1} term at that centre, even without varying the rest frame. □

Remark 5.6 (The same curvature coupling fixes the surface subtraction). The boundary coefficient has a gravitational counterpart, not merely a readout calibration. For the nonminimally coupled scalar Lagrangian

$$\mathcal{L} = \left(\frac{1}{16\pi G} - \frac{\xi}{2}\Phi^2\right)\mathcal{R} - \frac{1}{2}\sum_a (\nabla\phi_a)^2, \qquad \Phi^2 = \sum_a \phi_a^2,$$

the binormal curvature derivative gives the Wald surface density [20, Eqs. (53), (93)]

$$S_{\mathrm{W}}[\Sigma] = \int_\Sigma \left(\frac{1}{4G} - 2\pi\xi\Phi^2\right) dA.$$

The nonminimal scalar contribution and its relation to the entropy contact term are established results; see, for example, Donnelly and Wall [12, Eq. (5), Sec. II.A]. Here the same coefficient is used in the ball-family consistency law. Indeed, for the curvature coefficient c, $\partial\mathcal{L}/\partial R_{abcd} = c(g^{ac}g^{bd} - g^{ad}g^{bc})/2$; contraction with a binormal of norm -2 and multiplication by -2π give $4\pi c$. In the coherent witness the vacuum-subtracted quadratic profile is $\delta\langle\Phi^2\rangle = \theta^2\varphi_f^2$. At the retained order, with $h = O(\theta^2)$, its surface-measure correction is $O(\theta^4)$. For $\eta = 1/(4G)$, the Wald-matched ball quantity, with the same supplied volume term as F_B, is consequently

$$\mathcal{B}_B := \delta S_{\mathrm{W}}[\partial B] - \eta\frac{d-1}{R}\delta V(B) = \eta\,\delta F_B - \delta\langle Q_B\rangle.$$

Thus $\mathcal{B}_B + \delta\langle H_B\rangle = \eta\,\delta F_B + \delta\langle K_B\rangle$: the bulk consistency law is exactly restored. This matches a local surface density and fixes its matter coefficient from the same curvature coupling. It is a coefficient identity for the ball consistency law, not a thermodynamic first law for arbitrary diamonds.[1] The pure-area/raw-energy obstruction above is unchanged.

6 Which algebra the observer has

Consciousness I and Foundation I select an observer's algebra by noise and by a readable seed, and the selected algebra is in general smaller than the full local algebra [1, 4]. HAFF-D conjectures that gravitational phenomena are tied to the selection of observable algebras [3]. In the ball family the question is precise: does the geometry fixed by (MC_η) depend on which algebra is selected?

[1] An arbitrary ball edge is not a Killing horizon; MC, a generalized volume and a nonlinear second law do not follow from this density match.

For a bounded collection of scalar preparations and tested stress or collar smearings, choose a containing diamond D_0. The $U(1)$ Haar expectation E on this diamond preserves the vacuum and fixes the invariant readings. The semigroup $\mathcal{E}_t = e^{-\gamma t}\mathrm{id} + (1 - e^{-\gamma t})E$ is vacuum-stationary. Foundation I's stationary-source and stable-seed theorems [4] place the modular hull of any such invariant seed inside the proper fixed algebra. The charged coherent preparation above requires a separate phase-referenced source. A neutral quadratic seed supplies another preparation, entirely inside the same fixed algebra.

Proposition 6.1 (A neutral preparation inside a stable hull)**.** *In the free complex scalar theory of Proposition 5.3, let $X_a = \phi_a(f)$, $V_f = \omega(X_a^2) > 0$, and*

$$Q_f = \tfrac{1}{2}(X_1^2 + X_2^2), \qquad U_\theta = e^{i\theta Q_f}.$$

The strongly commuting fields X_1, X_2 define a self-adjoint Q_f by their joint spectral calculus. Both Q_f and U_θ are $U(1)$-invariant and local to any diamond containing supp f*. Include* arctan Q_f*, together with the desired bounded stress and collar readings, in the stable seed above. Its hull contains U_θ for every θ, while remaining inside the proper fixed algebra.*

Put $v_f = \varphi_f$, and define $H_f(x) = \frac{1}{2}\omega(\{X_a, \phi_a(x)\})$ for either fixed component $a = 1, 2$, with no sum over a; internal symmetry makes the two functions equal. Let $t[v]$ denote the quadratic tensor in brackets in (9)*, and polarize it as $t[v, z] = (t[v + z] - t[v] - t[z])/2$. The exact source is*

$$\delta\langle T\rangle_{\mu\nu} = 4\theta\, t_{\mu\nu}[v_f, H_f] + 2\theta^2 V_f\, t_{\mu\nu}[v_f]. \tag{12}$$

For every $\theta \neq 0$, it is smooth, conserved and traceless, has compact support on each spatial slice and has a finite Radon Fourier measure with all polynomial moments finite and nonzero weight away from zero. Its total energy is

$$\int \delta\langle T\rangle_{00}\, d^d x = \theta^2 V_f \int \left[(\partial_t v_f)^2 + |\nabla v_f|^2\right] d^d x > 0.$$

Its boundary datum is $\delta\langle\Phi^2\rangle = 4\theta v_f H_f + 2\theta^2 V_f v_f^2$.

The proof is in Appendix B.2.

The constraint construction of Example 5.4 gives $G^{\mathrm{lin}}[h] = (2\pi/\eta)\delta\langle T\rangle$, so this preparation identifies the canonical response. Its inputs are permission to apply the neutral quadratic pulse and the complete centre and frame family. Containing the stress tensor need not supply Q_f. Nor does neutrality by itself guarantee a nonzero tangent: the local conformal Wick square $:\Phi^2:(g)$ can have zero first-order vacuum stress response. The spatially smeared square used here is a different operator, and its positive finite-amplitude energy is computed explicitly. The noise fixes the invariant readings, although the prepared state need not be stationary.

There is a more general finite-energy route that uses a bounded observable already in the selected algebra. It needs no particular quadratic seed. Local source regularity remains a separate condition.

Proposition 6.2 (A bounded accessible preparation with positive energy)**.** *Let $H \geq 0$ be the translation generator, with* ker $H = \mathbb{C}\Omega$*. Let $\mathcal{N}$ be a selected von Neumann algebra for which Ω is separating. Suppose it contains a bounded non-scalar self-adjoint C, together with $\alpha_t(C) = e^{itH} C e^{-itH}$ for $0 < t < \tau$, where $\tau > 0$. For a sufficiently narrow real nonnegative $f \in C_c^\infty$ in that interval, $\int f = 1$, put*

$$B = \int f(t)\alpha_t(C)dt, \qquad U_\theta = e^{i\theta B}.$$

Then B is bounded, self-adjoint and non-scalar, $U_\theta \in \mathcal{N}$, and its vacuum excitation has finite energy for every θ. For every θ for which U_θ is non-scalar, in particular $0 < |\theta|\|B\| < \pi$,

$$0 < E_\theta := \langle U_\theta\Omega, HU_\theta\Omega\rangle \le |\theta|\|[H, B]\|.$$

With $K = [H, B]$, $\psi = (C - \langle\Omega, C\Omega\rangle)\Omega$ and the spectral measure μ_ψ of H, the quadratic coefficient is

$$\begin{aligned} E_2 &= \langle B\Omega, HB\Omega\rangle = \tfrac{1}{2}\langle\Omega, [B, [H, B]]\Omega\rangle \\ &= \int_{(0,\infty)} \lambda|\widehat{f}(\lambda)|^2\, d\mu_\psi(\lambda) > 0, \qquad E_2 \le \|B\|\|K\|, \end{aligned} \tag{13}$$

where $\widehat{f}(\lambda) = \int f(t)e^{it\lambda}dt$. An explicit remainder is

$$|E_\theta - \theta^2 E_2| \le 2|\theta|^3\|B\|^2\|K\| + \tfrac{1}{2}\theta^4\|B\|^3\|K\|.$$

If the state's vacuum-subtracted energy density is integrable on a time slice and its integral is E_θ, this gives a nonzero integrated source. Implementation permission and its cost are additional inputs.

The proof is in Appendix B.3.

The smoothed operator has bounded iterated commutators $\mathrm{ad}_H^m(B) = i^m \int f^{(m)}(t)\alpha_t(C)dt$. Repeated use of the product rule in the exponential series therefore gives $U_\theta\Omega \in \bigcap_m \mathrm{Dom}\, H^m$; all energy moments are finite. If μ_ψ has no atoms, narrowness is unnecessary: the nonzero entire function $\widehat{f}$ has only isolated zeros, which cannot carry that spectral measure. With atoms, a broad smearing can cancel an excitation exactly, so this refinement requires the stated spectral condition.

Corollary 6.3 (A common pulse inside a shared hull)**.** *Let several selected algebras $\mathcal{N}_j$ act in one vacuum representation with one positive generator H,* $\ker H = \mathbb{C}\Omega$*, and Ω separating for an ambient algebra containing them. Suppose each is invariant under* $\mathrm{Ad}\, e^{itH}$ *for $t \ge 0$. If their intersection $\mathcal{N}_\cap$ is non-scalar, it contains one common bounded pulse U_θ with finite strictly positive H-energy.*

Proof. The intersection is a von Neumann algebra, inherits separation and forward translation invariance, and contains a bounded non-scalar self-adjoint C. Apply Proposition 6.2 in $\mathcal{N}_\cap$. □

Proposition 6.4 (Finite algebras cannot carry this vacuum permission)**.** *Let $\mathcal{N}$ be finite-dimensional, let $H \ge 0$ have* $\ker H = \mathbb{C}\Omega$*, and suppose Ω is separating for $\mathcal{N}$. If* $\mathrm{Ad}\, e^{itH}(\mathcal{N}) \subseteq \mathcal{N}$ *for every $t \ge 0$, then $\mathcal{N} = \mathbb{C}1$. The ambient Hilbert space need not be finite-dimensional.*

Proof. Injectivity and finite dimension turn each inclusion into equality, so the automorphisms extend to negative times. Continuity fixes every minimal central projection. On each matrix block choose a self-adjoint generator K and matrix units E_{ab} in its eigenbasis: $\alpha_t(E_{ab}) = e^{it(k_a - k_b)}E_{ab}$. Since $H\Omega = 0$ and separation makes every nonzero $E_{ab}\Omega$ nonzero, this vector is an H-eigenvector with eigenvalue $k_a - k_b$. Positivity applied to E_{ab} and E_{ba} makes all differences zero. Thus the automorphisms fix $\mathcal{N}$, and $C\Omega$ lies in the one-dimensional kernel of H for every $C \in \mathcal{N}$. Separation then gives $C = c1$. □

The common pulse of Corollary 6.3 consequently uses an infinite field algebra when its full forward permission is exact. A finite observer can implement a finite-duration pulse or an explicitly controlled approximation; its controller Hamiltonian cannot simply be identified with the vacuum translation generator. The bounded-pulse construction requires only its stated finite interval and is unaffected.

Foundation I's Γ_3-hulls have precisely this forward translation closure [4, Theorem on Cut inheritance]. Thus a non-scalar shared intersection of those hulls, with the common representation hypotheses above, supplies a common energy-orientation test without adding Q_f. The positive generator must be the same for the readers: in a null Cut model it is null momentum, not automatically the time-translation Hamiltonian of this higher-dimensional theory. A merely modular-stable Γ_2-hull does not inherit translation permission. Realizing the pulse and reading its energy still require controls and an energy ledger.

There is a common physical-H instance in $3+1$ dimensions. For two independent free conformal complex scalar nets, take the future-cone algebra $\mathcal{A}(V_+)$ and the two selected nets $\mathcal{A}_1^{U(1)}\bar{\otimes}\mathcal{A}_2$ and $\mathcal{A}_1\bar{\otimes}\mathcal{A}_2^{U(1)}$. Time translations move V_+ into itself, the vacuum is separating there, and the physical H_1+H_2 has unique zero-energy vacuum [13, 14]. Each fixed algebra is taken on its own vacuum-cyclic subspace. The same physical Hamiltonian restricts to it because the Haar projection commutes with translations. Their intersection contains the neutral preparation of Proposition 6.1, the total stress and the invariant collar readings at each tested diamond. Thus the reference cone algebra and the selected ball family coexist in one matter model; they are distinct region algebras, not all operators inside one cone. The generic bounded pulse keeps its separate source-regularity condition. Shared implementation, calibrated collar limits and MC_η remain supplied permissions and consistency data.

Corollary 6.5 (A positive pulse supplies the spectral witness)**.** *Suppose a pulse from Proposition 6.2 has a smooth source $\delta\langle T\rangle$ in the tempered Radon Fourier class, and its vacuum-subtracted energy density is integrable on a time slice with integral $E_\theta > 0$. Then its source Fourier measure has nonzero weight away from $k=0$. If the response also belongs to that class and the full centre and open-frame consistency hypotheses of Theorem 4.4 hold, this source identifies the canonical response up to normalization.*

Proof. A Radon measure supported only at zero is a multiple of δ_0; the source would be constant. Its energy density is then constant on the infinite spatial slice. Integrability forces that component to vanish, contradicting $E_\theta > 0$. The identification follows from Theorem 4.4. □

This witness uses finite amplitude, not necessarily a nonzero first-order stress tangent. In fact $\partial_\theta E_\theta|_0 = i\langle\Omega,[H,B]\Omega\rangle = 0$ for every such vacuum pulse. A local tangent depends separately on $-2\,\mathrm{Im}\langle\Omega,T(x)B\Omega\rangle$. Proposition 6.1 computes both the regularity and a source explicitly; the general bounded construction does not supply Radon regularity from finite energy moments alone. Neither route establishes finite-resource identification of the entire geometric response.

Proposition 6.6 (Selection is invisible to bulk modular energy)**.** *For each ball let $\mathcal{A}_{\mathrm{sel}}(B)\subset\mathcal{A}(D(B))$ be a von Neumann subalgebra with a normal conditional expectation E_B that preserves the vacuum state, and suppose $\mathcal{A}_{\mathrm{sel}}(B)$ contains the bounded functions of stress-tensor smearings in $D(B)$; fixed-point algebras of a compact internal symmetry are examples. Then the modular group of $(\mathcal{A}_{\mathrm{sel}}(B),\omega)$ is the restriction of that of $\mathcal{A}(D(B))$, the quantity $\delta\langle K_B\rangle = 2\pi\int_B w_R\,\psi(T_{uu})$ is determined by the restriction of the state ψ to $\mathcal{A}_{\mathrm{sel}}(B)$, and (MC_η) is the same condition for every such choice. Every such selected algebra therefore fixes the same matter-sourced Einstein tensor. For $d\geq 3$, the metric additionally requires common initial*

or boundary data and is defined modulo gauge; for $d = 2$, it is locally fixed up to gauge on contractible domains. If the physical energy has an extra boundary term, the bulk datum must be read directly or that term must also be accessible and subtracted.

Proof. The first statement is Takesaki's theorem on conditional expectations [26]. The second holds because the stress-tensor smearings are affiliated with $\mathcal{A}_{\text{sel}}(B)$. The last follows from Theorem 4.2. □

Relative entropy behaves differently, as Foundation I found for current excitations of the SU(2) level-one net [4]. The same argument works in any dimension.

Proposition 6.7 (Selection changes entropy)**.** *Let $\mathcal{A}$ be the net of a free massless conformally coupled complex scalar $\phi = (\phi_1 + i\phi_2)/\sqrt{2}$ with its internal $U(1)$, let $\mathcal{A}_{\text{sel}}(B)$ be the $U(1)$-fixed points of $\mathcal{A}(D(B))$ and E_B the average over the group, and let $W = e^{i\theta\phi_1(f)}$, $\theta \neq 0$, where f is real, supported in $D(B)$, and has nonzero one-particle class. A nonzero test function alone would not suffice, since $f = \Box g$ can represent the zero field.*

(a) $S_{\mathcal{A}_{\text{sel}}(B)}(\omega_W \| \omega) < S_{\mathcal{A}(D(B))}(\omega_W \| \omega) = \delta\langle K_B \rangle_{\omega_W}$.

(b) As $\theta \to 0$, the modular energy is quadratic in θ, and the restricted state has vanishing first variation. At a faithful finite-dimensional regulator, write $\rho_\theta^{\text{inv}} = \rho_0 + \theta^2\chi + O(\theta^4)$. Then

$$S(\rho_\theta^{\text{inv}} \| \rho_0) = \frac{\theta^4}{2} g_{\text{BKM},\rho_0}(\chi, \chi) + O(\theta^6).$$

The continuum quartic law additionally requires a finite BKM tangent norm and uniform control of the entropy remainder; smooth field smearing and faithfulness alone do not establish these conditions.

The proof is in Appendix B.4.

A local coherent displacement preserves the regulated entropy of the full algebra. Its modular energy equals its relative entropy. The invariant algebra strictly loses relative entropy while retaining the stress charge; when the quartic expansion is controlled, its entropy increase exhausts that charge to order θ^2.

Appendix C computes the invariant mode coefficient and separates the exponential same-charge contribution from neutral-pair terms. A bounded BKM tangent norm is only one requirement: a continuum quartic entropy law still needs uniform control of the remainder as the regulators are removed. The strict entropy inequality above is independent of that expansion.

Conjecture 6.8 (Selection enters through second-order quantities or through the source)**.** *In the conformal ball setting, a change of the selected algebra at a fixed state changes the geometry reconstructed from ball modular data only through quantities of second order in the state perturbation, such as relative entropy, Fisher information and canonical energy, or through physical operations that change the stress tensor.*

Mechanism. The first-order datum is a function of the restriction of the state to stress-tensor observables, which every relevant algebra contains (Proposition 6.6). What sees more of the state is relative entropy, which depends on the algebra by monotonicity, and in holographic theories its second-order part for balls is the canonical energy that carries the second-order gravitational equations [28, 29].

Support. Propositions 6.6 and 6.7; Foundation I's strict inequality in $1 + 1$ dimensions [4]; the result of HAFF-D that a record alone does not move a gravitational source, while a preparation does [3].

Strongest objection. At second order the world still has one metric. The relative entropy of a smaller algebra measures what its observer cannot distinguish, so a second-order condition built from it may describe a coarser reconstruction rather than a different geometry. If so, the conjecture is a statement about the reach of an observer's reconstruction, which an observer-relative account of geometry has to quantify anyway, and not about a second spacetime.

Test. Compute $S_{\mathcal{A}(D(B))} - S_{\mathcal{A}_{\mathrm{sel}}(B)}$ for the charged coherent excitation of Proposition 6.7 through order θ^4, and compare the second-order consistency conditions built from the two relative entropies.

7 Second-order geometry and a source that follows it

The linear identity holds at every radius. At second order two of its inputs change: the intrinsic curvature need not equal twice the normal Einstein tensor, and the flat CHM charge need not be the modular Hamiltonian of the curved theory. We can resolve the first issue and identify the source terms needed for the second. This does not supply a nonlinear modular law.

7.1 A geometric two-jet consistency law

Put $c_0 = V_d/(d+2)$. Let g_ϵ be a smooth family of Lorentzian metrics with $g_0 = \eta$. At p, choose smooth spacelike hypersurfaces $\Sigma_{\epsilon,u}$ whose flat member is the hyperplane normal to u. Let n_ϵ be the future unit normal, q_ϵ its induced metric, and K_ϵ its second fundamental form. The family, including its centre and normal prescription, is part of the experiment. For the intrinsic geodesic ball of proper radius R, define

$$\begin{aligned} \mathfrak{F}_{R,u}(g_\epsilon) &= A_\epsilon - \frac{d-1}{R} V_\epsilon - \frac{\Omega_{d-1}}{d} R^{d-1} - \frac{c_0}{2} R^{d+1} \mathcal{Q}_\epsilon(p), \\ \mathcal{Q}_\epsilon &= (\mathrm{tr}_{q_\epsilon} K_\epsilon)^2 - K_{\epsilon,ij} K_\epsilon^{ij}. \end{aligned} \tag{14}$$

The last term is quadratic about a flat slice. It does not change any of the linear results, but it changes the second-order claim. This is a geometric correction dictated by the Gauss constraint, not an entropy or modular correction derived from a quantum state. The proper radius R and its multiplier $(d-1)/R$ are held fixed as ϵ varies. A coordinate sphere of unchanged coordinate radius is a different experiment: its geodesic radius and boundary measure change. The radial check below explicitly inverts that distance change.

For a smooth symmetric source T_ϵ, define a *stress reading*

$$\mathfrak{K}_{R,u}(T_\epsilon, g_\epsilon) = 2\pi \int_{B_\epsilon} \frac{R^2 - r^2}{2R} T_{\epsilon,\mu\nu} n_\epsilon^\mu n_\epsilon^\nu \, dV_\epsilon. \tag{15}$$

Here r is intrinsic geodesic distance, and the normal field on the slice is used inside the integral. In flat spacetime this is the bulk CHM charge. On a general curved metric it is only the specified local stress reading. No assertion of geometric modular flow on the curved diamond is being made.

Proposition 7.1 (Leading nonlinear ball identity)**.** *On a compact family with bounded smooth geometric and source derivatives, spacelike slices and radii below their normal injectivity radius,*

$$\mathfrak{F}_{R,u} = -c_0 R^{d+1} G_{\mu\nu}(g_\epsilon)(p) n_\epsilon^\mu n_\epsilon^\nu + O(R^{d+3}), \tag{16}$$

$$\mathfrak{K}_{R,u} = 2\pi c_0 R^{d+1} T_{\epsilon,\mu\nu}(p) n_\epsilon^\mu n_\epsilon^\nu + O(R^{d+3}). \tag{17}$$

The remainders and their first two ϵ derivatives are uniform when the corresponding mixed derivatives are bounded.

Proof. In intrinsic normal coordinates the volume density is $1-\mathrm{Ric}_{ij}\,x^ix^j/6+O(|x|^3)$. Angular averaging removes the odd terms and gives

$$V = V_d R^d \left(1 - \frac{{}^{(d)}\mathcal{R}}{6(d+2)} R^2\right) + O(R^{d+4}), \qquad A = \Omega_{d-1} R^{d-1} \left(1 - \frac{{}^{(d)}\mathcal{R}}{6d} R^2\right) + O(R^{d+3}).$$

Thus the area–volume difference minus its flat value is $-c_0 R^{d+1}\,{}^{(d)}\mathcal{R}/2 + O(R^{d+3})$. The exact Gauss equation ${}^{(d)}\mathcal{R} + \mathcal{Q} = 2G_{nn}$ proves the first statement. Taylor expansion of the smooth scalar T_{nn} and the volume density proves the second; its constant integral is $c_0 R^{d+1}$ and its odd terms average to zero. Differentiating these local expansions establishes the stated uniformity. Sufficient bounded mixed derivatives are assumed, rather than inferred from finite energy alone. □

Write, with no factorial in the second coefficient,

$$g_\epsilon = \eta + \epsilon h + \epsilon^2 j + O(\epsilon^3), \qquad T_\epsilon = \epsilon t_1 + \epsilon^2 t_2 + O(\epsilon^3), \tag{18}$$

$$G(g_\epsilon) = \epsilon G^{\mathrm{lin}}[h] + \epsilon^2 \big(G^{\mathrm{lin}}[j] + G^{(2)}[h,h]\big) + O(\epsilon^3). \tag{19}$$

$G^{(2)}$ is the quadratic coefficient of the covariant Einstein tensor in this fixed smooth identification of the metrics. Coefficients of the source use that same identification and the same renormalization convention. Appendix D gives its explicit coordinate formula and the source variation used below.

Theorem 7.2 (Second-order small-ball equivalence)**.** *Suppose $G^{\mathrm{lin}}[h] = (2\pi/\eta)t_1$ on an open domain and the two-jet families above have the regularity of Proposition 7.1. Then the following are equivalent:*

(a) At every p and for u in a nonempty open frame set,

$$\lim_{R\to 0} R^{-d-1}[\epsilon^2]\big(\eta\mathfrak{F}_{R,u} + \mathfrak{K}_{R,u}\big) = 0.$$

(b) The second-order tensor equation is

$$G^{\mathrm{lin}}[j] + G^{(2)}[h,h] = \frac{2\pi}{\eta} t_2.$$

Here $[\epsilon^2]$ denotes the quadratic Taylor coefficient. For one frame only the corresponding normal constraint follows.

Proof. Set

$$E(g_\epsilon) = G(g_\epsilon) - (2\pi/\eta)T_\epsilon.$$

Proposition 7.1 gives the leading residual $-\eta c_0 R^{d+1} E_{n_\epsilon n_\epsilon}$. Its zeroth and first coefficients vanish. Consequently changes of the normal at order ϵ multiply the vanishing first residual and do not alter the quadratic coefficient. That coefficient is $-\eta c_0 (G^{\mathrm{lin}}[j] + G^{(2)}[h,h] - (2\pi/\eta)t_2)_{uu}$. Tensor recovery from an open frame set proves both directions. □

This is a second-order conditional law in the same spacetime, not a holographic result. Unlike Theorem 4.2, it gives no exact finite-radius identity. Terms quadratic in curvature, its derivatives, the ball prescription and curved modular flow must be accounted for there. Nor does it classify all quadratic response functionals: adding terms whose small-ball leading coefficient vanishes leaves this test unchanged.

The extrinsic correction can instead be avoided by a particular physical region prescription. The local exponential image at p of $n_\epsilon^\perp$ is a spacelike geodesic hypersurface with $K_\epsilon(p) = 0$. On that family the uncorrected area–volume difference has the same leading law. It is not the same region as a ball on an arbitrary fixed laboratory slice. Regions, normals and preparations cannot be changed while interpreting their readings as the same experiment.

Example 7.3 (A bent slice in flat spacetime)**.** Keep the ambient metric exactly Minkowski and the source zero. Near the origin choose $t = \epsilon\alpha|x|^2/2$. Its normal there is ∂_t, and $K_{ij} = \pm\epsilon\alpha\delta_{ij} + O(\epsilon^2)$, so $\mathcal{Q} = \epsilon^2 d(d-1)\alpha^2 + O(\epsilon^3)$ and ${}^{(d)}\mathcal{R} = -\mathcal{Q}$. The raw ball difference is

$$[\epsilon^2]\left(F_R - \frac{\Omega_{d-1}}{d}R^{d-1}\right) = \frac{\Omega_{d-1}(d-1)}{2(d+2)}\alpha^2 R^{d+1},$$

although $G = 0$. To check the coefficient directly, its radial metric is $(1-\epsilon^2\alpha^2 s^2)ds^2 + s^2 d\Omega^2$; invert its proper-radius integral and expand its area and volume. The correction in Eq. (14) cancels this term exactly at quadratic order. Without a slice prescription or this correction, a nonlinear extension would assign a source to flat spacetime merely by bending a slice.

7.2 Why the flat prepared source cannot simply be reused

Even if a quantum stress expectation is smooth, a second-order response must use the total source in its perturbed metric. With a common covariant renormalization prescription and no gravitational anomaly, its Ward law is $\nabla^\mu T_{\mu\nu} = 0$. Expanding that identity gives

$$\partial^\mu t_{2,\mu\nu} = -\mathcal{D}_h[t_1]_\nu, \tag{20}$$

where

$$\begin{aligned}\mathcal{D}_h[t]_\nu &= -h^{\mu\alpha}\partial_\alpha t_{\mu\nu} - \eta^{\mu\alpha}\Gamma^{(1)\lambda}_{\alpha\mu}t_{\lambda\nu} - \eta^{\mu\alpha}\Gamma^{(1)\lambda}_{\alpha\nu}t_{\mu\lambda},\\ \Gamma^{(1)\lambda}_{\alpha\beta} &= \tfrac{1}{2}\eta^{\lambda\rho}(\partial_\alpha h_{\beta\rho} + \partial_\beta h_{\alpha\rho} - \partial_\rho h_{\alpha\beta}).\end{aligned}$$

Thus keeping only a conserved flat t_1 is generally not a second-order source: its curved divergence is $\epsilon^2\mathcal{D}_h[t_1]$. The expanded Bianchi identity demands precisely this correction.

The coherent preparation permits an explicit classical calculation of the missing terms. Take the same nonminimal scalar coupling as in Remark 5.6, now varied in the curved metric. For one real mean field Φ, its stress is

$$T_{\mu\nu}[g,\Phi] = \partial_\mu\Phi\partial_\nu\Phi - \tfrac{1}{2}g_{\mu\nu}(\nabla\Phi)^2 + \xi(g_{\mu\nu}\Box_g - \nabla_\mu\nabla_\nu)\Phi^2 + \xi G_{\mu\nu}(g)\Phi^2. \tag{21}$$

Its divergence is $(\Box_g\Phi - \xi\mathcal{R}_g\Phi)\partial_\nu\Phi$; the derivative of the last term cancels the curvature derivative from the improvement. Its trace vanishes on shell for the conformal value $\xi = (d-1)/(4d)$ at the classical level.

Set $\epsilon = \theta^2$ for the Weyl witness, and take $g = \eta + \theta^2 h_2 + \theta^4 h_4 + \cdots$ and $\Phi = \theta v + \theta^3 v_3 + \cdots$. Then

$$\Box v_3 = -\delta\Box_{h_2}v + \xi\mathcal{R}^{\text{lin}}[h_2]v, \tag{22}$$

$$t_2 = t[v], \qquad t_4 = 2t[v,v_3] + \delta_g t_{h_2}[v] + \xi G^{\text{lin}}[h_2]v^2. \tag{23}$$

Here $t_g[v]$ comprises the first three terms of Eq. (21); $\delta_g t_{h_2}[v]$ varies them with v fixed, and $t[v, z]$ is their flat polarization used in Proposition 6.1. In coordinates,

$$\delta\Box_h v = -h^{\mu\nu}\partial_\mu\partial_\nu v - (\partial_\mu h^{\mu\nu} - \tfrac{1}{2}\partial^\nu h^\alpha{}_\alpha)\partial_\nu v.$$

On any fixed finite causal domain with smooth h_2, the driven flat wave equation for v_3 has smooth solutions after its Cauchy data are specified. Taking zero correction data is one preparation prescription, not the same as keeping the field unchanged through the perturbed evolution. Equations (22)–(23) obey Eq. (20) at order θ^4. The curvature response, changed propagation and metric variation are calculable requirements, not optional control overhead. This is an actual two-jet source calculation within the supplied classical scalar action; it neither derives that action nor proves a quantum source apparatus realizes it.

This calculation has nonempty second-order content. Choose smooth compact Cauchy data with $v(p) = B \neq 0, \partial_t v(p) = A \neq 0$ and spatially constant data near p. They can be represented by a local Weyl preparation using a smooth time-slice cutoff. Then $t_{2,00}(p) = A^2/2 > 0$. Choose a constraint-compatible h_2 as in Example 5.4 and use a linearized normal-coordinate gauge at p, so $h_2(p) = 0$ and $\partial h_2(p) = 0$. Prescribe zero v_3 and first time derivative on that Cauchy slice. Tracefreeness gives $\mathcal{R}^{\text{lin}}[h_2](p) = 0$; Eq. (22) then gives zero second time derivative of v_3 at p as well. All second derivatives entering the polarized stress consequently vanish there, and $\delta_g t_{h_2}[v](p) = 0$. Therefore

$$t_{4,\mu\nu}(p) = \xi\kappa B^2 t_{2,\mu\nu}(p), \qquad t_{4,00}(p) = \xi\kappa B^2 A^2/2 > 0, \quad \kappa = 2\pi/\eta.$$

This is one supplied local preparation and coordinate prescription, not a gauge-invariant attribution to one separate term on every slice. The sum in Eq. (23) is the covariant source.

For a coherent displacement of a Hadamard reference on the *same* metric, with unchanged connected two-point function, point splitting gives exactly this state-dependent stress difference: the added two-point function is $\Phi(x)\Phi(y)$ and the state-independent counterterms cancel in the difference. The total gravitational source additionally contains the reference's renormalized stress, metric-dependent vacuum response and the preparation/control sources. Those terms cannot be declared zero by changing the reference independently on every ball. The scalar surface expectation and its measure also acquire order-θ^4 corrections. A Wald coefficient match remains useful, but does not turn a generic curved diamond into a Killing horizon or supply its modular Hamiltonian.

7.3 A strong common-geometry extension

Conjecture 7.4 (A causal source and a common second-order geometry)**.** *There is a physically preparable family in a common positive-energy covariant matter theory, with a fixed source reference and supplied gravitational data, for which the corrected small-ball stress consistency of Theorem 7.2 is the local limit of the actual modular/Wald response of its selected region family. It has the following additional properties:*

1. *The renormalized total source and its two-jet Ward identity include the controller, old reference, support and changed propagation, rather than only the outgoing mean pulse.*
2. *The geometric slice prescription and its quadratic extrinsic term are common to the observers. Their source readings and boundary subtractions identify one $G^{\text{lin}}[j] + G^{(2)}[h, h]$ with one normalization; changing a stress-containing algebra alone does not create a second sourced spacetime.*

3. *Selection changes relative-entropy reconstruction or changes the source through an actual intervention. A returned geometric record can then change a permitted preparation; deleting the record after emission does not undo its stress.*

4. *Finite-radius corrections have one causal source/geometry account, with separately controlled readout and collar errors. The small-ball limit is uniform in the prepared family, rather than obtained by spending unbounded resources on each ball independently.*

The flat Weyl and neutral pulses provide nonzero-frequency source witnesses for the first order; the scalar two-jet above provides an explicit propagation and Ward test for the next order. HAFF-D provides finite source–response–record returns in a supplied JT sector. Q-RAIF C provides an interacting quantum source with retained references and a common classical covariant action [6]. These are different models and dimensions. None yet proves this conjecture's common higher-dimensional quantum source or its actual curved modular response. The missing bridge is substantive: flat CHM covariance does not persist for arbitrary g, quantum reference stress can gravitate, and a finite positive regulator does not establish a renormalized covariant Ward law.

The bent-slice example is a geometric falsification test. A second is to insert the flat witness unchanged at order θ^4: unless $\mathcal{D}_{h_2}[t_2] = 0$ exceptionally, it fails the source Ward identity. A third compares charged and invariant-algebra relative entropies on one physical preparation while keeping the complete source fixed. Any resulting difference is a reconstruction difference unless a new source operation is demonstrated. These tests preserve the proposed connection between physical access and geometry without assigning gravitational stress to information that has merely been forgotten.

8 Higher multipoles and finite reading

Under a supplied radial Gaussian readout model, the ball coefficients give conditional information $\log I_k = -4k \log k + O(k)$. The resolved order consequently grows as $\log N/(4 \log\log N)$, as in Consciousness III's information-spectrum bound [2]. Appendix B.6 retains the normalized directions, noise assumptions, exact Schur complement and proof. This is a world-source reading model, not a self-description theorem.

9 Relation to other work, and limits

Entanglement equilibrium. Jacobson [19] compares, in a small geodesic ball, the variation of the area at fixed volume with the CHM modular energy, and obtains the Einstein equation from all rest frames at leading order; the coefficient $\Omega_{d-1}R^{d+1}/(d(d+2))$ is the c_0 of (24). The present paper adds the following: the fixed-volume coefficient is the fixed-reference variational multiplier and follows from full-kernel matching (Proposition 3.5); the comparison is exact at every radius (Theorem 4.2); among functionals of the intrinsic geometry, internal scales allowed, nonzero-frequency witnesses identify it up to normalization (Theorem 4.4); the condition has to be written with modular energy rather than entropy (Remark 4.8, Proposition 6.7); and the results say what an observer's partial data fix (Section 5).

The first law of causal diamonds. Jacobson and Visser [21] derive, from the covariant phase space of general relativity, a first law relating the variations of the edge area, the volume of the maximal slice, the cosmological constant and the matter Hamiltonian of a diamond in a

maximally symmetric space. Theorem 3.4 is its flat-space, linear geometric content, proved without a field equation.

Holography. Faulkner et al. [22] show that the first law for all boundary balls of a holographic CFT is equivalent to the linearized Einstein equation of the bulk; HAFF-D uses this for a supplied half-sided modular family [3]. There the geometry is the bulk geometry and the Ryu–Takayanagi formula supplies the geometric side [23]. Here the geometry is that of the spacetime in which the balls live, no dictionary is used, and (MC_η) takes the place of the Ryu–Takayanagi formula. The present result is a same-spacetime conditional route, with (MC_η) as its explicit hypothesis.

Non-conformal matter and higher curvature. For non-conformal fields the modular Hamiltonian of a ball is not the CHM charge, and Casini, Galante and Myers showed that for relevant deformations of low dimension the corrections dominate the small-ball limit [24]. The modular results and scalar preparations here use conformal matter; the leading two-jet geometric test only requires its stated smooth source. Higher-curvature versions of entanglement equilibrium use the Wald entropy and a generalized volume [25], consistent with Proposition 4.7(b).

Limits. The exact finite-radius results are linear in h about Minkowski space, with conformal matter and the specified bulk modular datum. Section 7 extends the local small-ball law to a geometric two-jet, with a separate stress consistency input. A physical boundary charge cannot be discarded when it varies; the scalar example makes both its compensating readout and the failure of the raw-energy law explicit. The constant η is supplied. (MC_η) is a hypothesis: the equivalence theorem identifies its field equation, and nonzero-frequency witnesses identify its intrinsic response up to normalization. They do not say why the modular energy of a ball should be balanced by a geometric response at all. Only balls are used.

10 Access, common geometry and the remaining mechanism

The ball family gives a precise sense in which different perspectives must agree with one geometry. Given the linear consistency condition, its intrinsic response is the Einstein tensor. A prepared nonzero-frequency source identifies that response; homogeneous or zero sources do not. The metric additionally needs gauge and common gravitational data. Comparison of rest frames recovers components inaccessible to a single frame, and finite rapidity or radius sets the precision of that recovery.

Stress-containing selected algebras share this sourced tensor. A coherent preparation loses relative entropy when restricted to an invariant algebra, while its stress remains. This distinguishes an observer's reach from a second physical geometry. The difference matters to Consciousness III's finite self-models without implying that a modular energy or a ball algebra is itself experienced [2].

The two-jet extension makes the next step concrete. An intrinsic ball alone can confuse a bent slice with a gravitational source; the Gauss correction removes that error. All small boosted balls then test one second-order residual, including the quadratic first-order geometry. The scalar source follows its perturbed propagation and curvature coupling. It cannot be held flat while its geometry is evolved. These are necessary parts of a causal selection–source–response mechanism, not its full derivation.

The central question is why the actual modular data obey the required geometric consistency at all. HAFF-D addresses physical selection and returned source response [3]; the present paper gives their higher-dimensional consistency and identification conditions. A common renormalized quantum source, curved modular response and controlled finite-radius law remain the high-value bridge in Conjecture 7.4. Curved backgrounds, non-conformal matter, the origin of η and shape-sensitive tidal readings remain further directions.

A Linear ball kernels and response identification

A.1 The small-ball series and exact Fourier transfer

Proposition A.1 (Small-ball series)**.** *Let $\delta\langle T\rangle$ be smooth near x_0. For every u and every N,*

$$\begin{aligned}\delta\langle K_B\rangle &= 2\pi \sum_{k=0}^{N} c_k\, R^{d+1+2k}\, \Delta_u^k\, \delta\langle T\rangle_{uu}(x_0) + O\big(R^{d+3+2N}\big),\\ c_k &= \frac{\Omega_{d-1}}{4^k\, k!\, (d/2)_k\, (d+2k)(d+2k+2)},\end{aligned} \tag{24}$$

with $\Omega_{d-1} = 2\pi^{d/2}/\Gamma(d/2)$ and $(a)_k$ the Pochhammer symbol. For a single Fourier component $\delta\langle T\rangle_{\mu\nu} = \mathrm{Re}(\varepsilon_{\mu\nu} e^{ik\cdot x})$, exactly,

$$\delta\langle K_B\rangle = \frac{2\pi V_d}{d+2}\, R^{d+1}\, f\big(R\,|k_u|\big)\, \mathrm{Re}\big(\varepsilon_{uu} e^{ik\cdot x_0}\big), \qquad f(z) = \Gamma\big(\tfrac{d}{2}+2\big)\Big(\frac{2}{z}\Big)^{d/2+1} J_{d/2+1}(z), \tag{25}$$

where $|k_u|^2 = k\cdot k + (u\cdot k)^2$ is the squared spatial momentum in the frame u, $V_d = \Omega_{d-1}/d$ and $f(z) = 1 - z^2/(2(d+4)) + O(z^4)$.

The proof is in Appendix A.2.

The leading coefficient is $c_0 = V_d/(d+2)$. For smooth inputs this is a finite-order asymptotic expansion; an infinite convergent series requires additional analyticity. The Fourier formula is exact. Section 8 uses these radial coefficients.

Corollary A.2 (Tensor recovery)**.** *The leading terms of* (24) *for all u in any nonempty open subset of the unit hyperboloid determine $\delta\langle T\rangle_{\mu\nu}(x_0)$.*

Proof. If $A_{\mu\nu}u^\mu u^\nu = 0$ on an open subset of the hyperboloid, it vanishes on all of it by analyticity, hence on the open cone of timelike vectors by homogeneity, so $A = 0$. □

A single frame gives only $\delta\langle T\rangle_{uu}$ at each point; Section 5.1 shows that this is a real loss, not an artefact of the leading order.

A.2 The small-ball coefficients

Proof. The spherical mean of a smooth function over the sphere of radius r about x_0 in Σ_u is $\sum_k r^{2k}\Delta_u^k g(x_0)/(4^k k!(d/2)_k)$ up to $O(r^{2N+2})$ (Pizzetti's formula [18]), and $\int_0^R \frac{R^2-r^2}{2R} r^{d-1+2k} dr = R^{d+1+2k}/((d+2k)(d+2k+2))$. For the Fourier component, $\int_{|y|<1}(1-|y|^2)e^{ip\cdot y}d^dy = \pi^{d/2}(2/p)^{d/2+1}J_{d/2+1}(p)$, and $e^{ik\cdot x}$ restricted to Σ_u is a plane wave of momentum $|k_u|$. □

A.3 Intrinsic distributional classification

Proof. Translate the centre to zero. By (F6) write $\delta F_B[h] = \sum_{ij}\langle L_{ij}, h_{ij}\rangle$, with a symmetric tensor distribution L supported in the closed ball. Gauge invariance gives $\partial_i L_{ij} = 0$, and rotation invariance makes $\sigma = \sum_i L_{ii}$ a radial distribution. A constant metric perturbation is pure gauge, so $\langle \sigma, 1\rangle = 0$.

Let Γ be the radial fundamental solution of the spatial Laplacian, $\Delta\Gamma = \delta_0$, and set $\mu_B = -\Gamma * \sigma/(d-1)$. This distribution is supported in the closed ball. Indeed, for $|x| > R$ the function $y \mapsto \Gamma(x-y)$ is harmonic on a neighbourhood of the ball. Its spherical average in y is the constant $\Gamma(x)$, by the mean-value property. Pairing with a radial distribution therefore gives $(\Gamma * \sigma)(x) = \Gamma(x)\langle\sigma, 1\rangle = 0$. The argument includes the logarithmic fundamental solution in $d = 2$.

For real $p \neq 0$, orthogonal covariance implies $\widehat{L}_{ij} = a(|p|)\delta_{ij} + b(|p|)p_i p_j$. Transversality and the trace then give

$$\widehat{L}_{ij}(p) = (|p|^2\delta_{ij} - p_i p_j)\frac{\widehat{\sigma}(p)}{(d-1)|p|^2} = (|p|^2\delta_{ij} - p_i p_j)\widehat{\mu}_B(p).$$

Both sides are Fourier transforms of compactly supported distributions, so equality extends through zero and identifies the distributions. Thus $L_{ij} = (\partial_i\partial_j - \delta_{ij}\Delta)\mu_B$. Distributional integration by parts proves (d) for arbitrary smooth h. The compact radial distribution μ_B has an even entire Fourier transform along any coordinate axis, which proves (a). Differentiation at zero proves (b). For (c), the Gauss equation ${}^{(d)}R + K^2 - K_{ij}K^{ij} = 2G_{\mu\nu}n^\mu n^\nu$ has $K_{ij} = O(h)$ about the flat slice, so at linear order $\tau_u = 2G^{\mathrm{lin}}_{uu}$. □

A.4 Fixed-reference multiplier and full kernel

Proof. (a) is a one-line differentiation. For (b), the functional depends on h only through τ_u (Lemma 3.2), so it can be evaluated on the conformally flat perturbation $h_{ij} = 2\phi\,\delta_{ij}$, for which $\tau_u = -2(d-1)\Delta\phi$. With the spherical means $\sum_k m_k s^{2k}$ of ϕ, the linearized geodesic ball gives $\delta A = (d-1)\Omega_{d-1}\sum_k m_k R^{d-1+2k}\frac{2k}{2k+1}$ and $\delta V = \Omega_{d-1}\sum_k m_k R^{d+2k}\frac{2k(d-1)}{(2k+1)(d+2k)}$; comparison with Proposition A.1 gives the ratio. □

A.5 The nonzero-frequency and Radon identification proof

Proof. On any open set of centres, the real and imaginary parts of a nonzero plane wave are linearly independent. Lemma 3.2(d) therefore gives

$$2\eta m_R(q_u)g_{uu} + 2\pi\widehat{w}_R(q_u)t_{uu} = 0, \qquad \widehat{w}_R(q) = \frac{V_d R^{d+1}}{d+2}f(Rq).$$

The entire function $\widehat{w}_R$ is nonzero. The transfer function m_R cannot vanish identically: otherwise t_{uu} would vanish on an open frame set and tensor recovery would give $t = 0$. Exclude the isolated zeros of m_R and $\widehat{w}_R$, and restrict to a frame patch whose q_u varies over an interval. There

$$g_{uu} = a(q_u)t_{uu}, \qquad a(q) = -\frac{\pi\widehat{w}_R(q)}{\eta m_R(q)}.$$

Lemma 4.3 gives $g = a_0 t$; the scalar is nonzero on this patch. The entire identity $2\eta a_0 m_R + 2\pi\widehat{w}_R = 0$ follows by analytic continuation. Set $c = 2\pi/(\eta a_0)$ to obtain the normalization. The real, reflection-invariant functional has a real transfer function on real momenta, so a_0 and

c are real. A compactly supported tensor distribution is determined by its Fourier transform, so equality on plane waves identifies the functional on all smooth inputs. The fixed nonzero amplitudes fix a_0, hence the same c, at every other tested radius.

For the measure extension, dominate all tensor components by the locally finite measure $\mu = \sum_{a,b}(|\widehat{\delta\langle T\rangle}_{ab}| + |\widehat{G}_{ab}|)$, and write their Radon–Nikodym densities as $t_{ab}(k), g_{ab}(k)$. Lemma 3.2(d) represents the centre dependence as convolution with a compact distribution on the slice. Its Fourier multiplier is $m_R(q_u(k))$, a smooth function of k whose real-axis growth is polynomial. Multiplication therefore preserves the tempered Radon class. Conservation holds for these densities almost everywhere. Fourier-transform consistency for a countable dense set of frames in the open frame set. Intersect their full-measure sets: at each remaining k, the displayed coefficient equation holds on the dense frame set, hence on the entire open set by continuity in u. The source hypothesis supplies one such $k \neq 0$ with $t(k) \neq 0$. The amplitude proof at that k identifies $m_R = -c\widehat{w}_R/2$.

Substitute this identity back into the measure equation. For almost every k, $\widehat{w}_R(q_u)[g(k) - (2\pi/(c\eta))t(k)]_{uu} = 0$ on the open frame set. If $k \neq 0$, q_u varies over an interval and the entire function $\widehat{w}_R$ has only isolated zeros. The tensor contraction therefore vanishes on an open frame patch and tensor recovery sets the bracket to zero. At $k = 0$, $\widehat{w}_R(0) > 0$ gives the same conclusion directly. Thus the tensor measures, and hence the distributions, obey the field equation. This uses densities with respect to a common measure, without dividing distributions. At any additional tested radius repeat the argument with that radius's own exceptional null set. The already established field equation fixes the same constant c, since the source is nonzero. No common full-measure set for uncountably many radii is required. □

A.6 The zero-frequency converse

Proof. Sufficiency is Theorem 4.4. A Radon measure supported at zero is a multiple of δ_0, so the remaining nonzero source is constant. In the common-measure proof above, its atom forces $m_R(0) \neq 0$. At almost every $k \neq 0$ the source density is zero and consistency gives $m_R(q_u)g_{uu} = 0$. Since m_R is nonzero and entire, varying the frame forces $g(k) = 0$. The response tensor is therefore constant as well. Adding $\ell^2 \int_B w_R \Delta_u \tau_u$, with $\ell \neq 0$, leaves these data unchanged but changes the transfer function by $-\ell^2 q^2 \widehat{w}_R(q)$, which is independent of the original transfer function because it vanishes at zero. □

B Prepared sources, bounded pulses and finite readings

B.1 The coherent source and its Radon measure

Proof. The Weyl commutator shifts the field by the classical solution $\theta\varphi_f$; normal ordering therefore gives (9) and $\delta\langle\sum_a :\phi_a^2:\rangle = \theta^2\varphi_f^2$. Since $\Box\varphi_f = 0$, differentiation gives conservation and zero trace. Finite propagation makes its Cauchy data compactly supported on every fixed time slice. The improvement in T_{00} is a spatial divergence, so

$$\int_{\mathbb{R}^d} \delta\langle T\rangle_{00}\, d^dx = \frac{\theta^2}{2}\int_{\mathbb{R}^d} \left[(\partial_t\varphi_f)^2 + |\nabla\varphi_f|^2\right] d^dx > 0.$$

Zero energy would make φ_f constant and, by compact spatial support, zero, contrary to the nonzero one-particle class.

Up to Fourier convention, the spacetime transform of φ_f is $\mathrm{sgn}(k^0)\delta(k^2)\widehat{f}(k)\, d^{d+1}k$. Its two mass-shell sheets carry densities proportional to $\widehat{f}(\pm|p|, p)\, d^dp/(2|p|)$. Near $p = 0$ the radial

total-variation integral is $O(r^{d-2}dr)$, integrable for $d \geq 2$; at infinity the restricted Schwartz function decreases faster than any power. Thus this is a finite complex measure with every polynomial moment finite. Derivatives multiply it by momenta. Products of these smooth fields have Fourier measures given by convolution; the improvement differentiates the convolution twice. These operations retain finite variation and every polynomial moment, proving the assertion for $\widehat{\delta\langle T\rangle}$. If that measure were supported only at zero, $\delta\langle T\rangle$ would be a constant tensor. Compact spatial support would force it to vanish, contradicting its positive total energy. Finally, a point on the edge of B is spacelike to every point in a compact subset of $D(B)$, so φ_f vanishes there and (10) is zero. □

B.2 The invariant quadratic source

Proof. The CCR and $[X_a, Q_f] = 0$ give the exact transformation $U_\theta^* \phi_a(x) U_\theta = \phi_a(x) + \theta v_f(x) X_a$. Applying it to the vacuum two-point function and the point-split stress tensor gives (12) and the boundary formula. Internal rotations preserve $X_1^2 + X_2^2$; injectivity of arctan on the spectrum implies that its bounded functional calculus contains U_θ. The stable-seed theorem then proves the hull assertions.

Both v_f and H_f are smooth real massless solutions. Their Fourier measures have the same rapidly decreasing mass-shell densities as in Proposition 5.3, with different phases on the two sheets. For $d \geq 2$ they have finite variation and every polynomial moment. The products and derivatives in (12) preserve these properties. Every term contains v_f or a derivative of it, so finite propagation gives compact spatial support. Polarization of the free-field identities proves conservation and zero trace.

The complex solution $\omega(X_a \phi_a(x))$ has only one frequency sign. Its real and imaginary parts have zero cross term in the integrated classical energy: in spatial Fourier variables the two amplitudes differ by multiplication by i. Since v_f is a real multiple of the imaginary part and H_f is its real part, $\int t_{00}[v_f, H_f]\, d^d x = 0$. The remaining energy is the displayed positive quadratic term. A Fourier measure supported only at zero would make the source constant, which contradicts compact spatial support and positive total energy. □

B.3 The bounded accessible pulse

Proof. The weak integral belongs to $\mathcal{N}$. Choose normalized nonnegative smearings whose supports shrink to a point $t_0 \in (0, \tau)$. Then $B\Omega \to \alpha_{t_0}(C)\Omega = e^{it_0 H} C\Omega$. Separation keeps $C\Omega$, hence this limit, off $\mathbb{C}\Omega$, so sufficiently narrow smearings remain non-scalar. Translation smoothing gives a bounded commutator $[H, B] = i \int f'(t) \alpha_t(C) dt$. Consequently U_θ preserves Dom H, with $\|[H, U_\theta]\| \leq |\theta| \|[H, B]\|$, and its vacuum excitation has finite energy. For $0 < |\theta| \|B\| < \pi$, the exponential is injective on the bounded spectrum of B, hence U_θ is non-scalar. If its energy were zero, the unique ground vector would give $U_\theta \Omega = z\Omega$, and separation would force $U_\theta = zI$, a contradiction. The upper bound follows from $E_\theta = \langle U_\theta \Omega, [H, U_\theta]\Omega \rangle$. The bounded commutator makes the exponential expansion valid in the graph norm of H. Since $H\Omega = 0$, the double commutator gives the quadratic term; the spectral theorem gives its integral and strict positivity. Put $R_\theta = (U_\theta - I - i\theta B)\Omega$. The exponential integral formula gives $\|R_\theta\| \leq \theta^2 \|B\|^2/2$ and $\|HR_\theta\| \leq \theta^2 \|B\| \|K\|$. Expanding the energy of $i\theta B\Omega + R_\theta$ gives the stated remainder. □

B.4 Strict entropy loss and the regulated coefficient

Proof. (a) The equality is (3); it is finite because a smooth coherent excitation has finite modular energy. Here the compactly supported preparation is strictly local to $D(B)$, so its surface variation (10) vanishes for this ball. Take a real test function g in $D(B)$ with nonzero commutator function $\Delta(f,g)$, and $x = \arctan \phi_1(g) \in \mathcal{A}(D(B))$. The rotation by π maps ϕ_1 to $-\phi_1$ and commutes with E_B, and arctan is odd, so $E_B(x) = 0$. In ω_W the field $\phi_1(g)$ has the vacuum Gaussian law shifted by $\theta\Delta(f,g) \neq 0$, so $\omega_W(x) \neq 0 = \omega_W(E_B(x))$. The state does not factor through E_B, and by Petz's criterion for sufficient subalgebras [27] the inequality is strict. (b) The coherent stress expectation is quadratic in its classical displacement. Rotation by π makes the restricted state even in θ. For a finite faithful density matrix, differentiating the logarithm gives

$$g_{\mathrm{BKM},\rho_0}(\chi,\chi) = \sum_{a,b} |\chi_{ab}|^2 \frac{\log p_a - \log p_b}{p_a - p_b},$$

with the diagonal value $1/p_a$; the first entropy derivative vanishes because $\operatorname{Tr}\chi = 0$. This proves the regulated assertion. □

B.5 Radial multipoles and finite reading

The higher terms of (24) are the radial multipoles of the local source, read through the modular weight. Consciousness III states a resolution order for any family of directions whose information falls as $e^{-\alpha j \log j}$, and notes that a physical instance needs normalized directions, a fluctuation kernel and a permitted measurement [2]. The ball family supplies the directions and the signal; the noise remains to be supplied.

Proposition B.1 (Fisher spectrum of the multipoles)**.** *Write the normalized reading of a ball as*

$$y(R) = \frac{\delta\langle K_B\rangle}{2\pi V_d R^{d+1}/(d+2)} = \sum_{k\geq 0} a_k \Big(\frac{R}{L}\Big)^{2k} \theta_k,$$

$$\theta_k = L^{2k}\Delta_u^k \delta\langle T\rangle_{uu}(x_0), \qquad a_k = \frac{d(d+2)}{4^k k!\,(d/2)_k\,(d+2k)(d+2k+2)},$$

exact when the analytic ball-average series converges on the radius range used. Suppose a record consists of n readings at radii drawn independently and uniformly from $(0,\rho L]$, each with an independent Gaussian error of variance σ^2. The Fisher information per record about θ_k, with $\theta_0,\dots,\theta_{k-1}$ unknown and all higher coefficients fixed and known (or with the series truncated at order k), is

$$I_k = \frac{n}{\sigma^2}\, a_k^2 \rho^{4k}\, \frac{1}{4k+1} \prod_{j<k} \frac{[2(k-j)]^2}{[2(k+j)+1]^2}, \qquad \log I_k = -4k\log k + O(k),$$

and ρ enters only at order k.

B.6 The radial Fisher Schur complement

Proof. The Fisher matrix is $F_{jk} = (n/\sigma^2)a_j a_k \rho^{2j+2k}/(2j+2k+1)$, a diagonally scaled Cauchy matrix with entries $1/(x_j+x_k)$, $x_j = 2j+\frac{1}{2}$. The information about θ_k with the lower parameters unknown is the last Schur complement of the leading $(k+1)\times(k+1)$ block, for which the Cauchy determinant gives the product. Stirling's formula gives $\log a_k = -2k\log k + O(k)$, and the product is $e^{O(k)}$. □

B.7 Numerical checks and applications

The closed form agrees with a Cholesky factorization at eighty-digit precision for $k < 14$ and $d = 2, 3, 4$. A readout weight comparable to a power of R, in the radius distribution or in the noise variance, changes the Schur complements by factors $e^{O(k)}$ and leaves the leading term. With Consciousness III's tolerance c, the number of multipoles resolved from N records is $K(N) \sim \log N/(4 \log\log N)$; for $d = 3$, $\rho = 1$ and $c = 1$ one finds $K = 8$ at $\log N = 100$ and $K = 55$ at $\log N = 1000$. The factorial suppression is a property of the modular weight, so the resolution exponent $\alpha = 4$ has a physical signal behind it. The readout model, including its noise, is supplied, and the directions belong to the world as read by the observer, not to the observer's model of itself.

C Regularity of the quartic entropy law

For a displaced charged thermal mode, put $z = e^{-b}$, $p_n = (1 - z)z^n$. Phase averaging gives the tangent $\chi_n = |\alpha|^2 p_n[(1 - z)^2 n/z - (1 - z)]$. The geometric variance $z/(1 - z)^2$ yields

$$g_{\mathrm{BKM}}(\chi, \chi) = \sum_n \chi_n^2/p_n = |\alpha|^4(1 - z)^2/z = 4|\alpha|^4 \sinh^2(b/2). \tag{26}$$

Thus finite modular energy $\sum_j b_j|\alpha_j|^2$ does not alone control the diagonal BKM coefficient.

For finitely many independent modes of one charge, the invariant coherences also contribute. With $m = n + e_j - e_l$, expansion of the displacement gives

$$\chi_{mn} = \alpha_j \overline{\alpha}_l \sqrt{(n_j + 1)n_l}\, p_n(1 - z_j)(1 - z_l)/z_l.$$

Using $p_m/p_n = z_j/z_l$ in the spectral BKM kernel and averaging $(n_j + 1)n_l$ gives

$$g_{\mathrm{BKM}}(\chi, \chi) = \sum_{j,l} |\alpha_j|^2|\alpha_l|^2(1 - z_j)(1 - z_l)\frac{\log z_j - \log z_l}{z_j - z_l}, \tag{27}$$

with coincident value $(1 - z_j)^2/z_j$. Opposite charges add the ordered neutral-pair sum with kernel $(1 - z_j)(1 - z_l)(b_j + b_l)/(1 - z_j z_l)$. The mean-value theorem bounds the same-charge kernel by $e^{b_j} + e^{b_l}$; the neutral-pair kernel is at most $b_j + b_l$. Consequently, writing $A = \sum_j |\alpha_j|^2$, $E = \sum_j e^{b_j}|\alpha_j|^2$ and $L = \sum_j b_j|\alpha_j|^2$, the full regulated tangent obeys $g_{\mathrm{BKM}} \le 2A(E + L)$. The full mode expansion is retained as `CII_BKM_mode_calculation.tex` in the supplementary source record.

These estimates control the tangent, not the entropy remainder. For the continuum ball net, a quartic law still requires its modular BKM norm and uniform remainder control as occupation, mode and volume regulators are removed. Compact smooth spacetime smearing has not been shown here to supply the exponential moment E. The exact strict relative-entropy inequality is independent of this open regularity question.

D Explicit two-jet geometry and source bookkeeping

All index moves in the following coefficients use η, and antisymmetrization is written out rather than hidden in a factor. For $g = \eta + \epsilon h + \epsilon^2 j$, the connection coefficients are

$$\Gamma^{(1)\lambda}_{\mu\nu}[h] = \tfrac{1}{2}\eta^{\lambda\rho}(\partial_\mu h_{\nu\rho} + \partial_\nu h_{\mu\rho} - \partial_\rho h_{\mu\nu}),$$
$$\Gamma^{(2)\lambda}_{\mu\nu}[j, h] = \Gamma^{(1)\lambda}_{\mu\nu}[j] - h^\lambda{}_\rho \Gamma^{(1)\rho}_{\mu\nu}[h].$$

Define the quadratic part with $j = 0$ by $\gamma_2 = \Gamma^{(2)}[0, h]$. Then

$$r^{(2)}_{\mu\nu}[h,h] = \partial_\lambda \gamma^\lambda_{2\,\mu\nu} - \partial_\nu \gamma^\lambda_{2\,\mu\lambda} + \Gamma^{(1)\lambda}_{\lambda\rho}\Gamma^{(1)\rho}_{\mu\nu} - \Gamma^{(1)\lambda}_{\nu\rho}\Gamma^{(1)\rho}_{\mu\lambda},$$
$$G^{(2)}_{\mu\nu}[h,h] = r^{(2)}_{\mu\nu} - \tfrac{1}{2}\eta_{\mu\nu}(\eta^{\alpha\beta} r^{(2)}_{\alpha\beta} - h^{\alpha\beta} R^{\text{lin}}_{\alpha\beta}[h]) - \tfrac{1}{2} h_{\mu\nu} R^{\text{lin}}[h].$$

These coefficients include the inverse-metric correction. The full quadratic Einstein coefficient is $G^{\text{lin}}[j] + G^{(2)}[h,h]$, not $G^{\text{lin}}[j]$ alone. For a first-order vacuum wave, the second-order vacuum equation is $G^{\text{lin}}[j] = -G^{(2)}[h,h]$; interpreting its right side as a gravitational energy tensor requires a gauge/averaging convention.

For the scalar define $s = v^2$ and

$$C^\rho_h = \eta^{\alpha\beta}\Gamma^{(1)\rho}_{\alpha\beta}, \qquad \delta\Box_h s = -h^{\alpha\beta}\partial_\alpha\partial_\beta s - C^\rho_h \partial_\rho s.$$

Varying the first three terms of Eq. (21) at fixed v gives the fully explicit quantity

$$\begin{aligned}\delta_g t_{h,\mu\nu}[v] = &-\tfrac{1}{2} h_{\mu\nu}(\partial v)^2 + \tfrac{1}{2}\eta_{\mu\nu} h^{\alpha\beta}\partial_\alpha v \partial_\beta v \\ &+ \xi\big(h_{\mu\nu}\Box s + \eta_{\mu\nu}\delta\Box_h s + \Gamma^{(1)\rho}_{\mu\nu}\partial_\rho s\big).\end{aligned}$$

At a normal-coordinate point this vanishes when h and its first derivatives vanish. The remaining $\xi G^{\text{lin}}[h]v^2$ does not, which is the nonzero-source calculation in Section 7. The identity $\nabla^\mu[(g_{\mu\nu}\Box - \nabla_\mu\nabla_\nu)s] = -R_{\nu\rho}\nabla^\rho s$, together with the Bianchi identity, reduces the full scalar stress divergence to its scalar equation times $\partial_\nu\Phi$. Its two-jet expansion gives Eq. (20), including the inverse metric and both connection terms. Counterterms cancel between two states only with one common metric and renormalization convention; the reference source remains in a total gravitational equation.

A finite algebraic check uses $g = e^{2\epsilon f}\eta$, $\Phi = \sqrt{\epsilon}e^{-(d-1)\epsilon f/2}v$, $f = t^2 + x^2/3 + tx/5$ and $v = 1 + 2t$. Direct rational-polynomial expansion through ϵ^2 in spacetime dimensions $3, 4, 5$ gives the conformal wave equation, vanishing trace and covariant divergence. Dropping $\xi G\Phi^2$ fails the divergence test. This tests the source identity on a prescribed conformal metric, not a solution of the coupled Einstein equations, and does not supply quantum renormalization or an actual curved modular Hamiltonian.

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